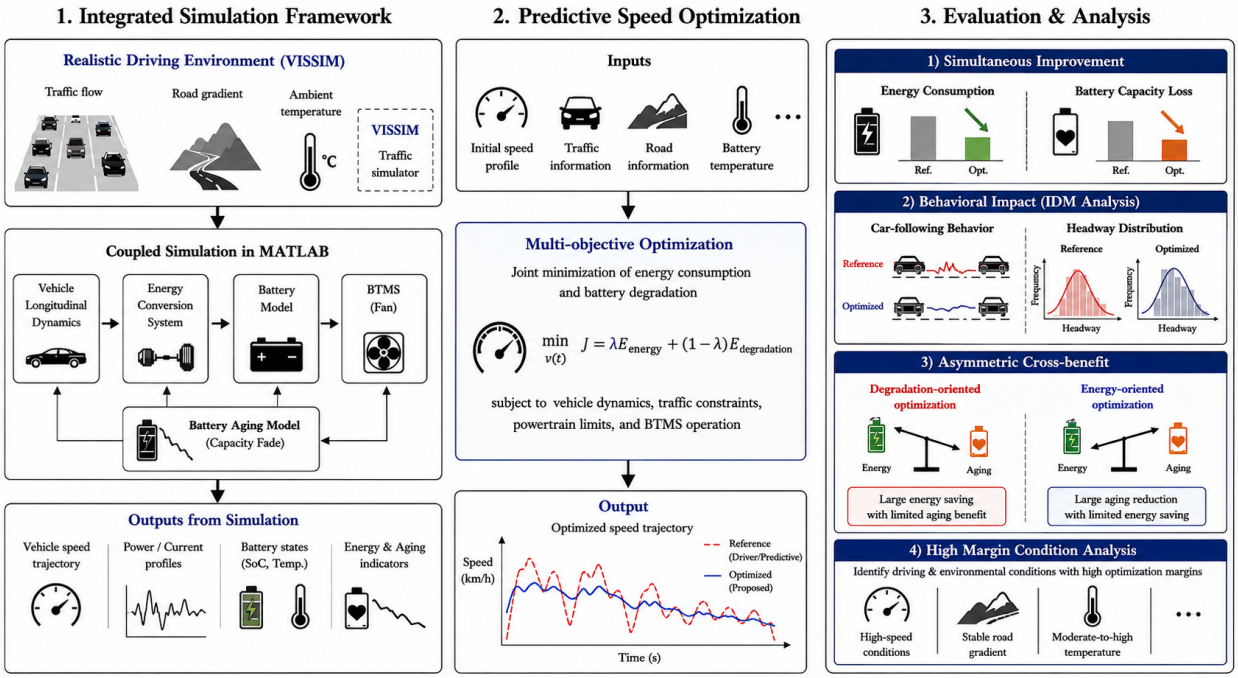
Graphical Abstract

Predictive Speed Control for Minimizing Energy Consumption and Battery Degradation in Battery Electric Vehicles: Analysis of High-Margin Driving Conditions

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Predictive Speed Control for Minimizing Energy Consumption and Battery Degradation in Battery Electric Vehicles: Analysis of High-Margin Driving Conditions

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Abstract

Battery electric vehicles (BEVs) require energy-efficient and degradation-aware control strategies under diverse real-world driving environments. Although predictive speed control has been widely investigated to improve driving efficiency, limited attention has been paid to the cross-objective relationship between energy consumption and battery degradation, the resulting changes in car-following behavior, and the operating conditions under which optimization provides the greatest benefits. To address these gaps, this study proposes a predictive speed control framework that jointly minimizes energy consumption and battery degradation under realistic traffic and road conditions.

An integrated simulation framework coupling microscopic traffic simulation, vehicle energy conversion, battery thermal management system (BTMS) operation, and battery aging models is developed using MATLAB and VISSIM. Optimal speed trajectories are determined under varying traffic flow, road gradient, and thermal conditions. The cross-objective benefits of energy consumption and battery degradation are evaluated across different objective weights, while the Intelligent Driver Model (IDM) is calibrated to characterize optimization-induced changes in car-following behavior. Hierarchical and interaction analyses are further conducted to identify the individual and conditional effects of driving and environmental factors on the achievable optimization margins.

The results demonstrate that predictive speed optimization reduces both energy depletion and battery capacity loss by suppressing aggressive acceleration and high-current events. The weighting-factor analysis further reveals an asymmetric cross-objective relationship: degradation-oriented optimization largely preserves the energy-saving benefits achieved by joint optimization, whereas energy-oriented optimization provides substantially smaller degradation-reduction benefits. The calibrated IDM and measured headways indicate that the optimized trajectories induce smoother and less reactive car-following behavior with increased following headways. Speed-related variables are the dominant determinants of optimization effectiveness, with larger margins primarily occurring under high-speed and stable-speed conditions. Road-grade variability also exhibits a significant conditional effect with average speed, demonstrating that optimization effectiveness depends on the combined driving and road conditions, whereas the thermal effect is comparatively limited. Overall, this study advances predictive speed control beyond aggregate performance evaluation by revealing the cross-objective, behavioral, and condition-dependent mechanisms underlying its benefits. The findings provide practical insights into how predictive speed optimization should balance energy efficiency and battery health and under which operating conditions it can deliver the greatest benefits for BEVs.

Keywords: Battery electric vehicle, Predictive speed control, Co-simulation framework, Multi-objective optimization, Driving condition analysis

Nomenclature

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Acronyms

ASM	Automotive simulation models
BEV	Battery electric vehicle
BT	Battery temperature
BTMS	Battery thermal management system
COP	Coefficient of performance of the cooling system
EOL	End-of-life
HVAC	Heating, ventilation, and air-conditioning system
IDM	Intelligent driver model
IR	Internal resistance
OCV	Open-circuit voltage
PCC	Pearson correlation coefficient
PTC	Positive temperature coefficient
SOC	State-of-charge

Variables

α	The first SOC dependence coefficient [-]
$\bar{v}$	Average vehicle speed [km/h]
β	The second SOC dependence coefficient [-]
β_{ij}	Coefficient for driving power conversion model [-]
Δv	Speed gap between ego vehicle and the leading vehicle [m/s] [-]
δ	Acceleration exponent for intelligent driver model [-]
Δt_s	Simulation resolution [s]
ϵ	C-rate dependence coefficient [-]
η_{heat}	Heating efficiency of PTC heater [-]
Γ	Battery life under current operating condition [Ah]
γ	Battery life under reference operating condition [Ah]
λ	Weighting factor for the objective function [-]
$\mathcal{A}$	Severity factor function [-]
$\mathcal{A}_{ref}$	Severity factor function evaluated at a nominal condition [-]
$\mathcal{D}$	dataset at optimization window[-]
$\overline{v}$	Upper bound of allowable speed range [m/s]
ρ_{regen}	Regenerative braking power ratio [-]

σ	Battery aging severity factor [-]
τ	Vehicle torque [Nm]
τ_{acc}	Vehicle torque for acceleration [Nm]
τ_{aero}	Vehicle torque for aerodynamic drag [Nm]
τ_{grad}	Vehicle torque for gradient resistance [Nm]
τ_{roll}	Vehicle torque for rolling resistance [Nm]
θ	Road slope [rad]
θ_{var}	Standard deviation of the relative road grade []
$\underline{v}$	Lower bound of allowable speed range [m/s]
$\varepsilon_{\text{dist}}$	Terminal distance tolerance [m]
ε_i	Regression coefficients for interaction regression model []
ζ	Optimization margin [Ah]
ζ_{Ah}^{trip}	Total energy depletion margin in a trip [%]
ζ_{loss}^{trip}	Total capacity loss margin in a trip [%]
A	Battery surface area [m^2]
a	Vehicle acceleration [m/s^2]
a_{max}	Maximum acceleration of ego vehicle [m/s^2]
Ah	Cumulative throughput of battery current [Ah]
b	Comfortable deceleration of ego vehicle [m/s^2]
C_{bat}	Specific heat capacity [J/(kg $\cdot$ K)]
C_{nom}	Nominal battery pack capacity [Ah]
d	Traveled distance of ego vehicle during simulation [m]
d_F, d_R	Front and rear spacing gaps [m]
d_{tol}	Minimum spacing tolerance [m]
d_{target}	Target travel distance [m]
E_a	Activation energy [$Jmol^{-1}$]
h	Convective heat transfer coefficient [W/(m$^2\cdot$K)]
I_{bat}	Battery demand current [A]
I_c	Battery pack C-rate [A/Ah]
J_{cons}	Normalized energy consumption cost [-]
J_{deg}	Battery degradation cost [-]
J_{tot}	Total cost [-]

k_{air} Aerodynamic drag coefficient [-]

K_{cool} Cooling gain parameter [W/K]

K_{heat} Heating gain parameter [W/K]

k_{RRC} Rolling resistance coefficient [-]

m Vehicle mass [kg]

m_{bat} Thermal mass [kg]

N_p Horizon window for optimization [-]

$N_{\|}$ Number of parallel cells in each module [-]

P_{req} Total battery power request [W]

P_{max} Maximum allowable battery power [W]

P_{bat} Total battery power [W]

P_{btm} Battery thermal management system power [W]

p_i Model coefficient for OCV-SOC equation [-]

P_{prop} Propulsion power [W]

P_{regen} Regenerative braking power [W]

P_{trac} Driving power [W]

Q_{max} Maximum allowable Joule heat generation [W]

Q_{btm} Heat flow required from BTMS [J]

Q_{env} Heat flow exchanged between battery and ambient environment [J]

Q_{EOL} End-of-life capacity loss [-]

Q_{gen} Heat generation rate by internal resistance of battery [J]

Q_{loss} Normalized capacity loss [%]

r_{gear} Gear ratio [-]

R_g Universal gas constant [$Jmol^{-1}k^{-1}$]

R_{in} Battery internal resistance [Ω]

R_{tire} Wheel radius [m]

s Distance (bumper-to-bumper) gap to the leading vehicle [m]

s^* Desired distance of ego vehicle [m]

s_0 Minimum distance (bumper-to-bumper) gap of ego vehicle [m]

s_F Relative distance from target vehicle to front vehicle [m]

s_R Relative distance from target vehicle to rear vehicle [m]

SOC SOC [%]

T	Absolute temperature [K]
T_{bat}	Battery temperature [K]
T_{env}	Ambient temperature [K]
t_h	Time headway to the leading vehicle [s]
T_{ref}	Target battery temperature [K]
v	Vehicle speed [m/s]
v_0	Desired speed of ego vehicle [m/s]
v_F, v_R	Front and rear vehicle speeds [m/s]
V_{bat}	Battery pack voltage [V]
V_{ocv}	Open-circuit voltage [V]
v_{var}	Standard deviation of vehicle speed [km/h]
v_{wind}	Wind speed [m/s]
z	Power-law exponent [-]

1. Introduction

Energy efficiency and battery degradation are key factors for the sustainable operation of battery electric vehicles (BEVs). Each factor directly affects vehicle usability and performance. For instance, low energy efficiency leads to excessive energy consumption under identical operating conditions, thereby increasing the risk of range anxiety before reaching the intended destination. Battery degradation reduces the total available energy and power capability of the battery system, resulting in degraded operational performance compared to the initial condition. Accordingly, extensive research efforts have focused on identifying environmental and driving factors that influence energy efficiency and/or mitigate battery degradation in BEVs.

Owing to the strong coupling among vehicle dynamics, battery operating conditions, and energy consumption in BEVs, energy efficiency and battery degradation have been widely investigated as key performance indicators. For example, energy efficiency has been shown to be strongly associated with the driving speed [1, 2, 3, 4] and the operation of the thermal management system [5, 6, 7]. In case of battery degradation, it has been reported to be influenced by various factors, including driving style [8, 9, 10], charging characteristics [8, 11, 12, 13], ambient temperature [8, 9, 13, 14, 15], and cumulative driving mileage [16, 17, 18, 19, 20].

The existing insights have primarily been obtained through impact assessment studies that investigate how energy efficiency and battery degradation in BEVs are influenced by various environmental and driving factors. Existing insights derived from impact assessment studies provide valuable prior knowledge on how environmental and driving factors affect energy efficiency and battery degradation in BEVs. Such knowledge plays an important role in guiding the formulation of predictive control problems, for example, in the design of objective functions, constraints, and previewed variables.

The predictive control problems can be broadly categorized into power-split strategy and speed control approaches for energy management. Power-split strategies typically formulate optimization problems to determine the optimal power distribution between the battery and other onboard power sources, with the objective of improving energy efficiency [21, 22, 23, 24, 25] and, in some cases, explicitly mitigating battery degradation of the target vehicle [26, 27, 28, 29, 30]. However, these approaches are inherently applicable only to electrified vehicle systems equipped with multiple onboard power sources (e.g., hybrid or fuel-cell electric vehicles), which limits their direct applicability to pure BEVs. On the other hand, the speed control method is applicable to all types of electric vehicles, regardless of their powertrain configuration. Table 1 summarizes the analysis scope of optimization problems addressed within predictive speed control frameworks. The table highlights a clear trend toward incorporating speed-level and road dynamic conditions to directly optimize vehicle speed trajectories with the objective of minimizing energy consumption and/or mitigating battery degradation.

In the early stage, optimal driving guidance frameworks were primarily developed to compute energy-efficient speed profiles based on prior knowledge of road geometry, such as altitude and slope information [31, 32]. These studies established the fundamental concept of speed optimization using previewed road information. In the subsequent stage, optimization problems were further developed to determine the optimal speed profile of an ego vehicle under facility-related constraints, with a primary focus on minimizing driving energy consumption in signalized traffic environments [33, 34, 35, 36]. More recently, predictive speed control frameworks have evolved to explicitly incorporate traffic conditions. Specifically, these optimization approaches have been extended beyond isolated signalized scenarios to account for vehicle interactions and surrounding traffic dynamics, thereby enabling predictive speed control under more realistic and complex traffic environments [37, 38, 39, 40]. In addition, several studies have begun to incorporate battery degradation considerations into predictive speed control formulations, further extending their objectives beyond energy consumption minimization [41, 42].

Accordingly, the next stage of research calls for a unified optimization framework that can comprehensively address both energy efficiency and battery degradation under diverse and dynamic operating conditions. Such a framework is essential for systematically capturing the coupled effects of driving behavior, environmental conditions, and traffic dynamics on energy consumption and battery degradation; however, it remains largely unexplored in the existing literature [45]. Moreover, most existing predictive speed control studies have primarily focused on low- to moderate-speed driving scenarios, such as urban or signalized traffic environments. Although high-power operating conditions—particularly high-speed driving—are known to induce both increased energy consumption and accelerated battery degradation [8, 10, 14], limited attention has been devoted to optimizing speed profiles that explicitly and jointly consider energy efficiency and battery degradation in complicated road dynamic environments.

Table 1: The overview of previous optimization problem studies for predictive speed control in BEVs

Paper	High-speed driving condition	Road dynamic condition			Objective function	
		Facility	Geometry	Traffic	Energy efficiency	Battery degradation
Yi and Bauer. (2015) [32]	O	X	O	X	Driving	X
Wu et al. (2015) [33]	X	O	O	X	Driving	X
He and Wu. (2018) [34]	X	O	X	X	Driving	X
Li et al. (2018) [35]	X	O	X	X	Driving	X
Morlock et al. (2019) [36]	O	O	O	X	Driving	X
Kang et al. (2020) [37]	O	O	X	O	Driving	X
Zhou et al. (2022) [43]	X	X	X	O	Driving	X
Jia et al. (2022) [41]	X	O	X	O	X	O
Yeom et al. (2022) [38]	O	X	O	O	Driving	X
Sun et al. (2023) [44]	O	X	X	O	Driving	X
Su et al. (2024) [42]	X	X	X	O	Driving	O
Pan et al. (2025) [39]	X	O	O	O	Driving	X
Chen and Cao. (2025) [40]	X	O	X	O	Driving	X
This work	O	O	O	O	Driving	O

Although several studies have incorporated energy consumption and battery degradation into speed optimization problems, limited attention has been paid to the cross-objective benefits arising from their joint optimization. Since both objectives are physically coupled through vehicle dynamics and battery operating conditions [46], prioritizing one objective may also improve the other [42]; however, such cross-objective benefits are not necessarily proportional or reciprocal [41, 47]. This raises an important question as to whether degradation-oriented optimization can preserve energy-saving benefits to the same extent that energy-oriented optimization preserves degradation-reduction benefits.

Moreover, existing studies have primarily focused on aggregate performance improvements, while comparatively less attention has been paid to how optimized speed trajectories alter the underlying driving behavior and how the resulting benefits vary across driving and environmental conditions. In particular, the car-following characteristics induced by predictive speed optimization and the conditional effects among traffic dynamics, road geometry, and thermal conditions on the achievable optimization margins remain insufficiently understood. Clarifying these relationships is therefore essential not only for determining whether and how energy efficiency and battery degradation should be jointly considered, but also for explaining why predictive speed optimization is effective and identifying the combinations of operating conditions under which its benefits can be maximized.

Therefore, this paper aims to address the aforementioned research gaps by developing a unified predictive speed optimization that integrates energy consumption and battery degradation models. The proposed framework formulates a multi-objective optimization problem to determine optimal speed trajectories under diverse and dynamic operating conditions. To reproduce realistic traffic interactions, the framework is implemented within the VISSIM traffic simulation environment, where the ego vehicle interacts with surrounding vehicles during the optimization process. Beyond evaluating the overall optimization performance, this study investigates the cross-objective relationship between energy efficiency and battery degradation, characterizes the resulting car-following behavior, and identifies the individual and conditional effects of traffic, road geometry, and thermal conditions on the achievable optimization margins. The main contributions of this paper are summarized as follows.

1. A unified predictive speed optimization framework is developed to jointly improve energy efficiency and mitigate battery degradation under diverse traffic, road, and thermal conditions.
2. The cross-objective relationship between energy efficiency and battery degradation is quantitatively investigated to identify their coupled and asymmetric optimization benefits.
3. The optimized car-following behavior is characterized to reveal how predictive speed control modifies ego-vehicle interactions with surrounding traffic.
4. The individual and conditional effects of traffic dynamics, road geometry, and thermal conditions are systematically analyzed to identify the operating conditions associated with high optimization margins.

The remainder of this paper is organized as follows. Section 2 describes the construction of the integrated simulation framework and associated vehicle and battery models. Section 3 presents the proposed predictive speed optimization method for jointly minimizing energy consumption and battery degradation under dynamic traffic conditions. Section 4 evaluates the optimization performance and further investigates the cross-objective relationship between energy efficiency and battery degradation, the resulting car-following behavior, and the individual and conditional effects of driving and environmental factors on the achievable optimization margins. Finally, Section 5 concludes the paper by summarizing the main findings and contributions and outlining directions for future research.

2. Model Construction

We constructed models for the predictive speed control framework. Figure 1 illustrates the overall simulation framework. In MATLAB, preview vehicle-level information are obtained from the VISSIM traffic simulation through the COM interface. Energy consumption and battery degradation are then evaluated using the integrated powertrain and battery aging models. The framework further enables speed control of the target vehicle within the traffic simulation, with the control inputs determined through MATLAB-based optimization. Due to the high complexity of powertrain dynamics in BEV systems, directly solving optimization problems using high-fidelity models is computationally challenging. Therefore, many studies have adopted simplified powertrain models, provided that their essential characteristics and accuracy are sufficiently preserved for control-oriented analysis.

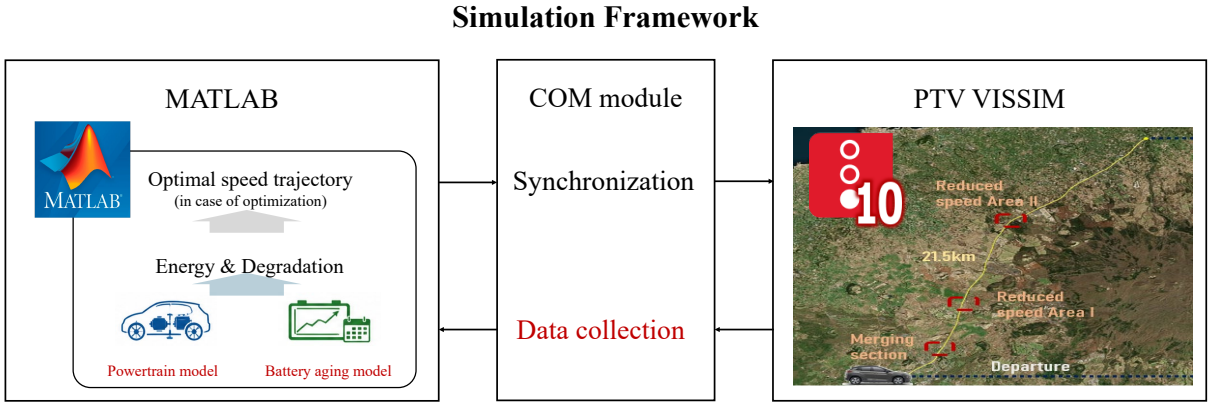


Figure 1: The architecture of traffic-integrated vehicle driving simulation framework for speed control-based optimization

2.1. Powertrain Model

For tractable optimization, the powertrain model must be simplified while preserving its dominant physical characteristics. In BEV systems, battery power demand primarily arises from driving loads and thermal management operations [7]. Thermal management power consists mainly of heating, ventilation, and air-conditioning (HVAC) and the battery thermal management system (BTMS). Unlike HVAC loads, which are subject to significant uncertainties due to passenger comfort requirements and ambient conditions, BTMS operation is more systematically controlled based on battery thermal states. Therefore, this study focuses on only power components that can be modeled deterministically within the optimization framework, excluding highly uncertain auxiliary loads. Accordingly, the analysis considers driving power and battery thermal management power as the primary contributors to battery energy consumption. The total battery power, P_{bat}, is therefore defined as the sum of driving power, P_{trac}, and battery thermal management power, P_{btm}.

$$P_{bat} = P_{trac} + P_{btm} \tag{1}$$

The driving power, P_{trac}, is systematically modeled using system-level variables, such as vehicle speed and acceleration, based on established driving energy consumption models for BEV systems [1, 21, 48, 49, 50, 51]. Accordingly,

a control-oriented simplified model, including propulsion power (P_{prop}) and regenerative braking power (P_{regen}), is adopted in this study, in which vehicle torque (τ) and speed (v) are used as primary inputs to characterize the driving power, as expressed below.

$$P_{trac} = \begin{cases} P_{trac,prop} = \beta_0 + \sum_{i=0}^{3}\sum_{j=0}^{3}\beta_{ij}\tau^i v^j \\ P_{trac,regen} = \rho_{regen}\left[e^{\left(\frac{0.0411}{|a|}\right)}\right]^{-1} \cdot P_{trac,prop} \end{cases} \tag{2}$$

The coefficient, β_{ij}, was identified using dSPACE Automotive Simulation Models (ASM) as a high-fidelity vehicle simulation platform. The regenerative braking power ratio, ρ_{regen}, was experimentally set to 0.125 based on calibration results obtained from the ASM simulator. Here, $|a|$ denotes the magnitude of vehicle acceleration. The regenerative braking model is adopted from [50]. This platform provides a realistic representation of power flow and efficiency in BEV systems across diverse operating conditions. Using the electrified powertrain model implemented in dSPACE ASM, time-series data from each component were collected through vehicle-level simulations and subsequently used to construct the energy consumption model. To reduce computational cost in optimization problem, we simplified the electrified powertrain model as a polynomial form in 2. Based on the collected datasets, a simplified battery power model was developed with high R^2 value, as shown in Figure 2.

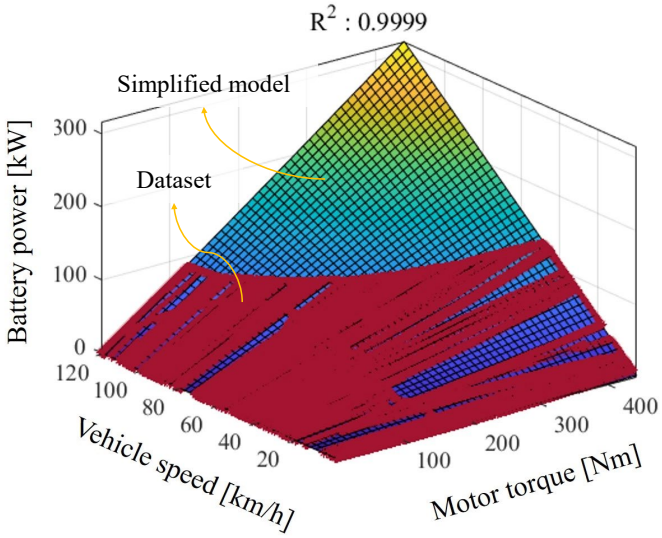


Figure 2: The simplified battery power model with vehicle speed and motor torque

The motor torque required to overcome the forces acting on the vehicle can be derived from a longitudinal vehicle dynamics model. Specifically, the total motor torque is expressed as the sum of individual torque components associated with acceleration, aerodynamic drag, road gradient, and rolling resistance, as follows:

$$\tau = \tau_{acc} + \tau_{aero} + \tau_{grad} + \tau_{roll} \tag{3}$$

$$\tau_{acc} = \frac{maR_{tire}}{r_{gear}} \tag{4}$$

$$\tau_{aero} = \frac{k_{air} \cdot (v + v_{wind})^2 R_{tire}}{r_{gear}} \tag{5}$$

$$\tau_{grad} = \frac{mg\,sin\theta R_{tire}}{r_{gear}} \tag{6}$$

$$\tau_{roll} = \frac{k_{RRC} mgcos\theta R_{tire}}{r_{gear}} \tag{7}$$

where the motor torque τ is composed of the acceleration torque (τ_{acc}), aerodynamic drag torque (τ_{aero}), gradient resistance torque (τ_{grad}), and rolling resistance torque (τ_{roll}). Here, m denotes the total vehicle mass, a is the vehicle acceleration, R_{tire} represents the wheel radius, and r_{gear} is the gear ratio. The parameter k_{air} denotes the aerodynamic drag coefficient, v is the vehicle speed, and v_{wind} represents the wind speed. The road slope is denoted by θ, and k_{RRC} represents the rolling resistance coefficient.

2.2. Battery thermal model

In the BTMS, the thermal dynamics of the battery module are approximated using a simplified lumped thermal model with thermal mass m_{bat} and specific heat capacity C_{bat} [52, 53, 54, 55, 56]:

$$\dot{T}_{bat}(t) = \frac{1}{m_{bat} C_{bat}} \left(\dot{Q}_{gen} + \dot{Q}_{env} + \dot{Q}_{btm} \right), \tag{8}$$

with

$$\dot{Q}_{gen} = I_{bat}^2 R_{in}, \tag{9a}$$

$$\dot{Q}_{env} = hA \left(T_{env} - T_{bat} \right), \tag{9b}$$

where $\dot{Q}_{gen}$ represents the heat generation due to the internal resistance of the battery, and T_{bat}, I_{bat}, and R_{in} denote the battery temperature, current, and internal resistance, respectively. Moreover, $\dot{Q}_{env}$ denotes the heat flow exchanged between the battery and the ambient environment, where T_{env} is the ambient temperature, and h and A represent the convective heat transfer coefficient and the battery surface area, respectively. The required BTMS heat flow, $\dot{Q}_{btm}$, generated by the compressor and battery pump through the battery-coupled chiller loop during cooling, or by the positive temperature coefficient (PTC) heater during heating, is determined by a rule-based BTMS controller [57] to regulate the battery temperature within the desired operating range as follows:

$$\dot{Q}_{btm} = \begin{cases} -\mathrm{K}_{cool} \left(T_{bat} - T_{ref} \right), & \text{if } T_{bat} > T_{ref}, \\ \mathrm{K}_{heat} \left(T_{ref} - T_{bat} \right), & \text{if } T_{bat} \leq T_{ref}, \end{cases} \tag{10}$$

where T_{ref} denotes the desired battery temperature, and K_{cool} and K_{heat} denote the cooling and heating gain parameters equal to 450 and 350, respectively. The power consumption of the BTMS is modeled as a linear function of the BTMS heat flow rate in (10) [53, 55] as

$$P_{btm} = \begin{cases} -\dfrac{\dot{Q}_{btm}}{\mathrm{COP}}, & \dot{Q}_{btm} < 0 \\ \dfrac{\dot{Q}_{btm}}{\eta_h}, & \dot{Q}_{btm} \geq 0 \end{cases} \tag{11}$$

where COP denotes the coefficient of performance of the cooling system equal to 1.5, and η_h represents the heating efficiency of the PTC heater equal to 2.8.

2.3. Battery electrical model

In the battery electrical model, the battery current is related to the total power demand, which consists of the traction power demand (P_{trac}) in (2) under the given driving conditions and the BTMS power consumption (P_{btm}) in (11), as follows:

$$I_{bat} = \frac{P_{trac} + P_{btm}}{V_{bat}}, \tag{12}$$

where the battery terminal voltage, V_{bat}, is represented by a simplified internal resistance model as follows:

$$V_{bat} = V_{ocv}(\text{SOC}) - I_{bat}R_{in}(T_{bat}, \text{Ah}), \tag{13}$$

where SOC and Ah represent the state of charge and cumulative ampere-hour throughput of the battery, respectively, and V_{ocv} and R_{in} denote the open-circuit voltage (OCV) and internal resistance of the battery, respectively. Assuming that the battery terminal voltage varies slowly over the driving cycle, $V_{bat}[k]$ is approximated by its previous-step value in the discrete-time, i.e., $V_{bat}[k] \approx V_{bat}[k-1]$. Accordingly, the battery current can be calculated as

$$I_{bat}[k] = \frac{P_{trac}[k] + P_{btm}[k]}{V_{bat}[k-1]}. \tag{14}$$

Using the battery current calculated in (14), the SOC dynamics are described by the Coulomb-counting equation as

$$\dot{\text{SOC}}(t) = -\frac{I_{bat}(t)}{C_{nom}}, \tag{15}$$

where C_{nom} denotes the nominal battery capacity. The SOC trajectory is then obtained by integrating (15):

$$\text{SOC}(t) = \text{SOC}(t_0) - \frac{1}{C_{nom}} \int_{t_0}^{t} I_{bat}(\tau) d\tau. \tag{16}$$

Based on the updated SOC, battery temperature, and cumulative ampere-hour throughput (Ah) at the current time step, the OCV (V_{ocv}) and the internal resistance (R_{in}) are calculated from their corresponding empirical relationships. Then, the battery terminal voltage, V_{bat}, is updated using (13) and utilized as the previous-step voltage for calculating the battery current at the next time step.

As the SOC dynamically varies dynamically during driving scenarios, the SOC is estimated using an OCV–SOC relationship. In this study, the OCV–SOC relationship is approximated by a polynomial model derived from experimental data obtained from a real battery module, as described in Equation (17) [58]. The battery module used for data acquisition was manufactured for BEV applications by LG Chem, and its specifications, along with those of the test equipment, are detailed in Table 2.

$$V_{ocv} = p_0 + p_1 x + p_2 x^2 + p_3 x^3 + p_4 x^4 + p_5 x^5 + p_6 x^6, x = \frac{(Ah - \mu_1)}{\mu_2} \tag{17}$$

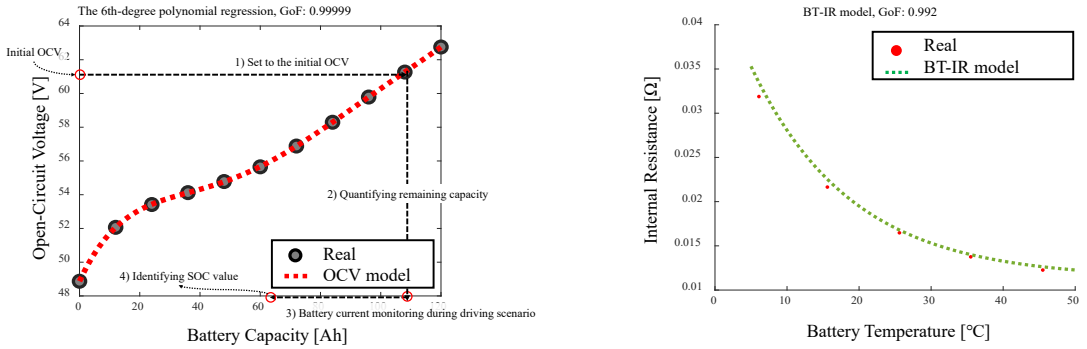
Table 2: The specifications of the real battery module, battery cycler, and thermal chamber

Battery module specification		Battery cycler and chamber specification	
Battery cell	LG Chem E63B	Output voltage	~ 500V
Architecture	2P15S	Output current	-240 ~ 240 A
Cell type	NMC811 & Pouch	Data acquisition	10 ms ~
Voltage range	40.5 ~ 63 V	Temperature range	-40 ~ 100 °C
Nominal capacity	120Ah	Humidity range	20 ~ 98 %RH

The polynomial OCV–SOC model was constructed in MATLAB R2024b using the polyfit function. As shown in Figure 3a, the fitted model accurately captures the OCV–SOC relationship, achieving a goodness-of-fit (GoF) of 0.999. The estimated coefficients are as follows: $p_0 = 55.6712$, $p_1 = 3.5195$, $p_2 = 1.7234$, $p_3 = -0.2774$, $p_4 = -0.6877$, $p_5 = 0.3320$, and $p_6 = -0.0190$. It should be noted that the Ah variable was normalized using its mean and standard deviation prior to model fitting, where $\mu_1 = 60$ denotes the mean and $\mu_2 = 39.7995$ denotes the standard deviation. Using the polynomial OCV-SOC model, the SOC is estimated throughout the driving scenario. Initially, the SOC is determined from the OCV at the start of the trip, which serves as the reference for remaining capacity. Subsequently, the SOC is updated in real time by applying the battery current demand using a coulomb-counting approach during the driving process. Moreover, since internal resistance varies with battery temperature, an exponential model is adopted to characterize the battery temperature–internal resistance (BT–IR) relationship, as described in Equation (18)

[59, 60, 61, 62, 63, 64]. As shown in Figure 3b, the fitted model accurately captures the BT–IR relationship, achieving a goodness-of-fit (GoF) of 0.992. The estimated coefficients are $a = 0.0343$, $b = 0.0717$, and $c = 0.0113$. In this study, the OCV and BT-IR models are scaled by a factor of six to reflect the configuration of a real BEV battery pack, thereby extending the voltage range to 243–372 V [65, 66].

$$R_{in} = (a \cdot e^{-bT} + c) \tag{18}$$



(a) (b)

Figure 3: Battery modeling based on real battery module data: (a) sixth-order polynomial OCV–SOC model with coulomb-counting-based SOC estimation process; (b) exponential battery temperature–internal resistance (BT–IR) model.

2.4. Battery Aging Model

Based on the powertrain and BTMS models, the total battery power demand is determined for given driving scenarios. In optimization-based frameworks, semi-empirical aging models have been widely adopted to quantify battery degradation under varying power demands and operating conditions [67, 68, 69]. In this study, we employ a representative semi-empirical aging model in 19, which is well-suited for integration into optimization problems due to its balance between physical interpretability and computational efficiency.

$$Q_{loss} = (\alpha \cdot SOC + \beta) \cdot e^{\left(\frac{-E_a + \epsilon I_c}{RT}\right)} \cdot Ah^z \begin{cases} \alpha = 1287.6, \beta = 6356.3 \; at \; SOC \leq 0.45 \\ \alpha = 1385.5, \beta = 4193.2 \; at \; SOC > 0.45 \end{cases} \tag{19}$$

where Q_{loss} denotes the normalized capacity loss, α and β characterize the dependence on SOC, which is defined differently for SOC levels below and above 45%. Here, SOC represents the instantaneous state of charge, and E_a is the activation energy equal to 31,700 $[Jmol^{-1}]$. The parameter ϵ describes the dependence on the C-rate equal to 163.3, where I_c denotes the C-rate. In addition, R_g is the universal gas constant equal to 8.314 $[Jmol^{-1}k^{-1}]$, and T represents the absolute temperature. The term Ah denotes the cumulative charge throughput of the battery current, and z is the power-law exponent that governs the dependence on Ah throughput equal to 0.57.

The capacity loss model in (19) describes battery degradation as a function of SOC, C-rate, temperature, and cumulative charge throughput; following the severity factor map definition in [68], the aging factor in this model is defined as the severity factor function evaluated under the current battery operating condition:

$$\mathcal{A} = (\alpha \cdot SOC + \beta) \cdot e^{\left(\frac{-E_a + \epsilon I_c}{R_g T}\right)}, \tag{20}$$

For a given end-of-life (EOL) capacity loss threshold Q_{EOL}, the cumulative charge throughput required to reach the EOL condition under the current operating condition and the reference operating condition are expressed as

$$\gamma = \left(\frac{Q_{\text{EOL}}}{\mathcal{A}}\right)^{1/z}, \quad \Gamma = \left(\frac{Q_{\text{EOL}}}{\mathcal{A}_{\text{ref}}}\right)^{1/z}, \tag{21}$$

where $\mathcal{A}_{\text{ref}}$ denotes the severity factor function evaluated at a nominal reference operating condition. The reference condition is selected as $SOC_{\text{ref}} = 0.5$, $I_{c,\text{ref}} = 1.0$, and $T_{\text{ref}} = 298.15$ K, which provides a consistent baseline for comparing the relative severity of different battery operating conditions. Accordingly, the degradation severity factor, corresponding to the severity factor map in [68], is defined as the ratio between the reference equivalent life and the equivalent life under the current operating condition:

$$\sigma = \frac{\Gamma}{\gamma} = \left(\frac{\mathcal{A}}{\mathcal{A}_{\text{ref}}} \right)^{1/z}. \tag{22}$$

A larger σ indicates a more severe aging condition for the same amount of battery current throughput. This degradation severity factor is used to formulate the battery degradation cost in the optimization objective (see Section 3 for details).

2.5. Vehicle Driving Simulation Framework

To integrate vehicle-level driving simulation with traffic simulation, the VISSIM traffic simulator (Vissim 7) is coupled with MATLAB (R2024b) via a COM interface. Figure 4 illustrates the study area located along Pyeonghwa road (21.5 km) on Jeju Island. Pyeonghwa Road is an arterial roadway connecting Jeju City and Seogwipo City, with no signalized intersections along the corridor. In the simulation environment, traffic congestion rarely occurred except near merging sections. To introduce variability in driving conditions, two reduced-speed zones were implemented at 6,500 m and 13,500 m along the roadway. The first zone has a length of 350 m, while the second extends for 300 m. Within these zones, reduced desired speeds were applied: 30 km/h for passenger vehicles and 25 km/h for heavy vehicles. The study corridor features an uphill segment in the first half and a downhill segment in the second half, as it traverses Mt. Halla from Seogwipo City toward Jeju City. The elevation profile was generated by linearly interpolating altitude data at each GPS point obtained from the Google Maps API.

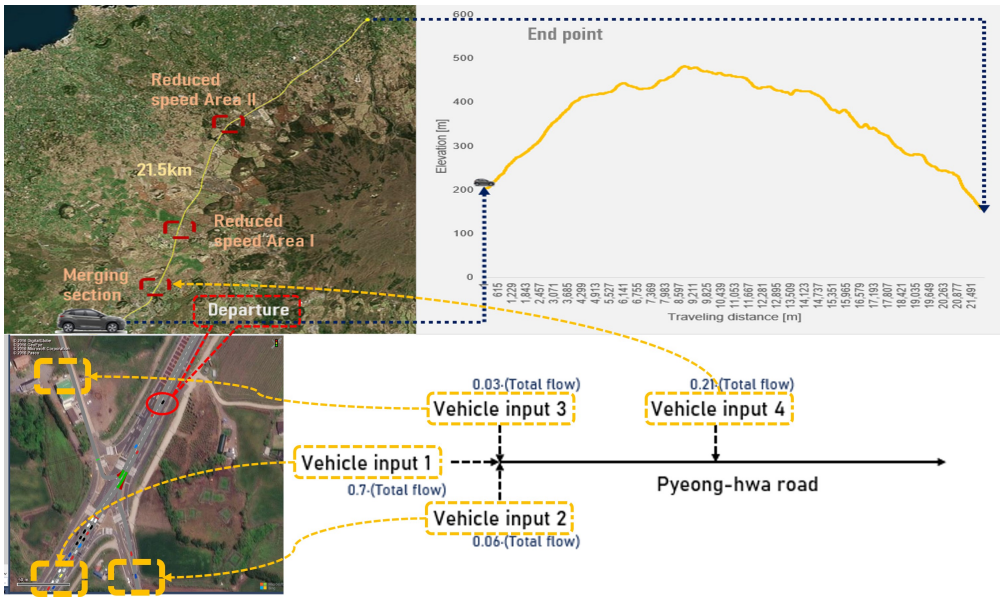


Figure 4: Vehicle driving simulation configurations with road gradient, road infrastructure, and traffic flow

Within the simulation framework, vehicle flow rates are controlled via vehicle input objects in VISSIM. Four input locations were defined across the study corridor, including the Seogwang intersection and a merging section located in the middle of the roadway. The input flow rates were derived from field traffic observations collected along Pyeonghwa Road.

To capture varying traffic conditions, multiple demand scenarios were considered, with total flow rates of 1,800, 3,600, and 5,400 vehicles per hour. To ensure that each scenario reaches a steady-state traffic condition, a warm-up

period of at least 900 seconds is applied at the beginning of the simulation [70, 71, 72, 73]. Following the warm-up phase, a vehicle is randomly selected at initial departure point from the traffic stream and used for the subsequent vehicle-level driving simulation. In addition, ambient temperatures of 0, 10, 20, 30, and 40°C are considered. These conditions influence the variation of BTMS power during each trip, thereby enabling the analysis of thermal effects on both energy consumption and capacity degradation. During the simulation, driving information of the target vehicle—including speed, acceleration, roadway position, and headway distance to the leading vehicle—is transferred to MATLAB via a COM interface. The road gradient is obtained from a predefined gradient map and evaluated at the vehicle's position using linear interpolation between discrete roadway points.

3. Simulation-based Optimization Problem Formulation

The goal of the proposed simulation-based speed trajectory optimization framework is to derive an optimized speed profile that reduces battery energy consumption and mitigates battery degradation while maintaining travel progress comparable to the reference driving trajectory. Battery degradation is affected not only by battery current and C-rate but also by battery temperature [68], while BTMS operation for battery temperature regulation introduces additional energy consumption [7]. Therefore, the speed trajectory optimization is formulated as a multi-objective problem that considers traction energy, thermal management energy, and battery degradation characteristics. In this optimization framework, the preview data required for the optimization problem are constructed from the reference driving trajectory generated in the VISSIM simulation:

$$\mathcal{D}_k = \left\{ d_{\text{target},k}(i),\ d_{F,k}(i),\ d_{R,k}(i),\ v_{F,k}(i),\ v_{R,k}(i) \right\}_{i=1}^{N_p}, \tag{23}$$

where $\mathcal{D}_k$ denotes the preview data set at the current optimization window k, and N_p represents the prediction horizon length. At the i-th prediction step of the k-th window, the reference VISSIM trajectory provides the target travel distance, the longitudinal positions of the front and rear vehicles, and the speeds of the front and rear vehicles, denoted by $d_{\text{target},k}(i)$, $d_{F,k}(i)$, $d_{R,k}(i)$, $v_{F,k}(i)$, and $v_{R,k}(i)$, respectively.

3.1. Objective Function

The optimization problem determines the speed trajectory over the prediction horizon N_p to reduce battery energy consumption and battery aging while satisfying the physical constraints imposed by the powertrain and BTMS models. For the k-th optimization window, the objective consists of the normalized battery energy consumption cost, J_{cons}, and the battery degradation cost, J_{deg}. The energy consumption cost represents the battery energy used for vehicle traction and BTMS operation, while the degradation cost penalizes aging-related operating conditions associated with battery current and temperature. Accordingly, the total cost is defined as

$$J_{\text{tot},k}(\mathbf{v}_k) = \sum_{i=1}^{N_p} \left[\lambda J_{\text{cons},k}(i) + (1-\lambda) J_{\text{deg},k}(i) \right], \tag{24}$$

where λ is the weighting factor that balances energy consumption and battery degradation, and $\mathbf{v}_k$ denotes the speed variables to be optimized over the prediction horizon. The normalized energy consumption cost is calculated from the traction power and BTMS power computed by the powertrain and BTMS models as

$$J_{\text{cons},k}(i) = \frac{w_{trac} P_{trac,k}(i) \Delta t_s / 3600 + w_{btm} P_{btm,k}(i) \Delta t_s / 3600}{V_{nom} Q_{nom}}. \tag{25}$$

where w_{trac} and w_{btm} are the weighting parameters for the traction and BTMS energy terms, respectively, Δt_s is the sampling time step, and V_{nom} and Q_{nom} denote the nominal voltage and capacity of the battery. The battery degradation cost is calculated as

$$J_{\text{deg},k}(i) = \frac{\sigma_k(i) |I_{cell,k}(i)|}{I_{cell,\text{ref}}}, \tag{26}$$

where $\sigma_k(i)$ is the degradation severity factor obtained from (22), and $I_{cell,\text{ref}}$ is the reference cell current used for normalization.

3.2. Optimization Problem with Traffic Constraints

To optimize the speed trajectory of the target vehicle while satisfying the distance, speed, and acceleration constraints with respect to the surrounding vehicles over the prediction horizon (i.e., $i = 1, \ldots, N_p$), the acceleration and traveled distance of the ego vehicle are recursively computed as

$$\begin{aligned} a_k(i) &= \frac{v_k(i) - v_k(i-1)}{\Delta t_s}, \\ d_k(i) &= d_k(i-1) + v_k(i-1)\Delta t_s, \end{aligned} \tag{27}$$

where $v_k(0)$ and $d_k(0)$ denote the initial speed and position of the target vehicle at the beginning of the k-th optimization window, respectively. Using the preview data in (23), the relative distances and relative speeds with respect to the front and rear vehicles are calculated at each prediction step as

$$\begin{aligned} s_{F,k}(i) &= d_{F,k}(i) - d_k(i), \quad & s_{R,k}(i) &= d_k(i) - d_{R,k}(i), \\ \Delta v_{F,k}(i) &= v_k(i) - v_{F,k}(i), \quad & \Delta v_{R,k}(i) &= v_{R,k}(i) - v_k(i), \end{aligned} \tag{28}$$

where $s_{F,k}(i)$ and $s_{R,k}(i)$ are the relative distances from the target vehicle to the front and rear vehicles, respectively, and $\Delta v_{F,k}(i)$ and $\Delta v_{R,k}(i)$ are the corresponding relative speeds. The speed bounds are then obtained from the front and rear vehicle distance constraints as

$$\overline{v}_k(i) = v_{F,k}(i) + \frac{s_{F,k}(i) - d_{\text{tol}}}{\Delta t_s}, \quad \underline{v}_k(i) = v_{R,k}(i) + \frac{d_{\text{tol}} - s_{R,k}(i)}{\Delta t_s}. \tag{29}$$

with the safety distance tolerance d_{tol}.

The optimization problem combines the battery energy consumption and degradation costs with the collision-avoidance constraints for the front and rear vehicles. Including the physical constraints of the powertrain and BTMS models, the problem over the prediction horizon N_p is formulated as follows:

$$\mathbf{v}_k^* = \arg\min_{\mathbf{v}_k \in \mathcal{V}_k} \; J_{\text{tot},k}(\mathbf{v}_k) \tag{30a}$$

$$\text{s.t.} \quad (8), (15),$$

$$\underline{a}_k(i) \le \frac{v_k(i) - v_k(i-1)}{\Delta t_s} \le \overline{a}_k(i), \tag{30b}$$

$$\underline{v}_k(i) \le v_k(i) \le \overline{v}_k(i), \tag{30c}$$

$$s_{F,k}(i) \ge d_{\text{tol}}, \tag{30d}$$

$$s_{R,k}(i) \ge d_{\text{tol}}, \tag{30e}$$

$$P_{trac,k}(i) \le P_{trac,\max}, \tag{30f}$$

$$\dot{Q}_{gen,k}(i) \le Q_{\max}, \tag{30g}$$

$$P_{btm,k}(i) \le P_{btm,\max}, \tag{30h}$$

$$|d_k(N_p) - d_k^{ref}(N_p)| \le \varepsilon_{\text{dist}}. \tag{30i}$$

where $\mathbf{v}_k = [v_k(1), \ldots, v_k(N_p)]^\top$ denotes the speed sequence to be optimized, and $\mathcal{V}_k$ is the set of admissible speed sequences in the k-th optimization window. The optimal speed sequence $\mathbf{v}_k^*$ is obtained by solving the resulting constrained nonlinear programming (NLP) problem using the SQP algorithm in `fmincon` from the MATLAB Optimization Toolbox. After solving the problem in MATLAB, the optimized speed sequence is applied to the target vehicle in VISSIM, and the next optimization window is generated, as illustrated in Figure 5.

4. Result and Discussion

In this section, the optimization results are systematically investigated under diverse traffic, thermal, and driving environmental conditions. The effectiveness of the proposed predictive speed control framework is evaluated by

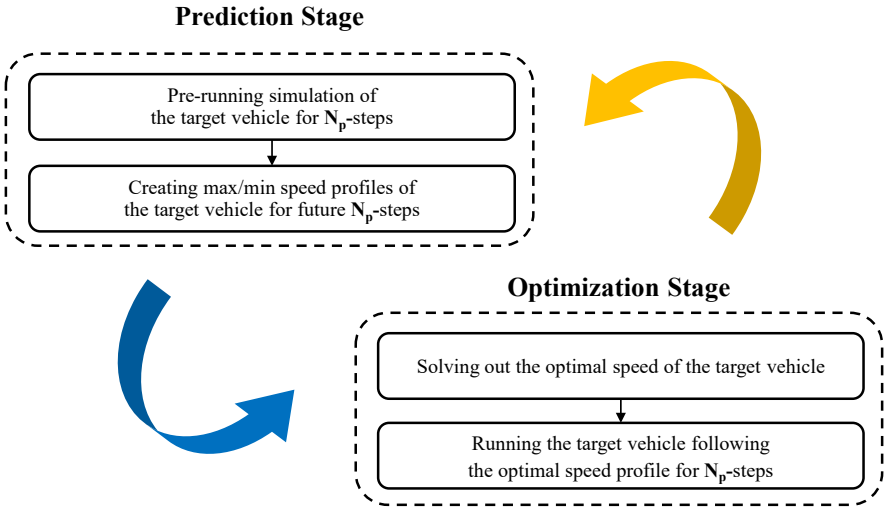


Figure 5: Simulation-based speed optimization framework using the integrated MATLAB-VISSIM simulation

comparing the reference and optimized driving profiles in terms of energy consumption and battery degradation. The analysis further examines how traffic flow and ambient temperature influence the optimization performance.

Figure 6 demonstrates that the proposed traffic simulation-based framework generates feasible predictive speed trajectories under realistic traffic interactions. Existing predictive speed optimization methods generally assume that future traffic evolution can be adequately represented by predicted vehicle trajectories. While this assumption is reasonable under simple car-following scenarios, it becomes increasingly restrictive in realistic traffic environments where traffic states continuously evolve due to interactions among multiple surrounding vehicles. As a result, the optimized speed trajectory may no longer remain feasible or optimal under complex traffic conditions. The proposed framework explicitly incorporates interactions between the ego vehicle and surrounding traffic, including cut-in events, lane changes, and overtaking maneuvers.

The two graphs in Figure 6 illustrate that the ego vehicle's leading vehicle can change as a result of dynamic traffic interactions. Consequently, both the relative speed and the inter-vehicle distance continuously vary over time. These traffic interactions continuously influence the feasible speed adjustment range and are therefore integrated into the optimization process. Consequently, the optimized driving profile improves both energy efficiency and battery degradation performance while maintaining realistic traffic behavior and safe interactions with surrounding vehicles. The following sections first examine the driving characteristics of the optimized profile, then present the optimization results under various traffic and environmental conditions, and finally identify the driving conditions associated with high optimization margins.

4.1. Comparison between reference and optimized profiles

~~A representative trip segment is randomly selected to compare the reference and optimized driving profiles.~~ Figure 7 presents a zoomed-in view to highlight the differences between the reference and optimized profiles. Over each prediction horizon of N_p steps (10 seconds), the pre-recorded (reference) speed profile is replaced by the optimized trajectory. For the subsequent horizon, the initial speed is set to the final value of the optimized profile from the previous step, ensuring continuity across successive optimization intervals. As a result of the optimization, the acceleration profiles are noticeably smoothed compared to the reference case, with reduced fluctuations and lower peak magnitudes, thereby contributing to improved energy efficiency and mitigated battery degradation. This result is consistent with driving behavioral analysis in previous studies [42, 74].

To provide a quantitative comparison of driving behavior between the reference and optimized driving profiles, the Intelligent Driver Model (IDM) in (31) was employed. The calibrated IDM parameters enable numerical interpretation

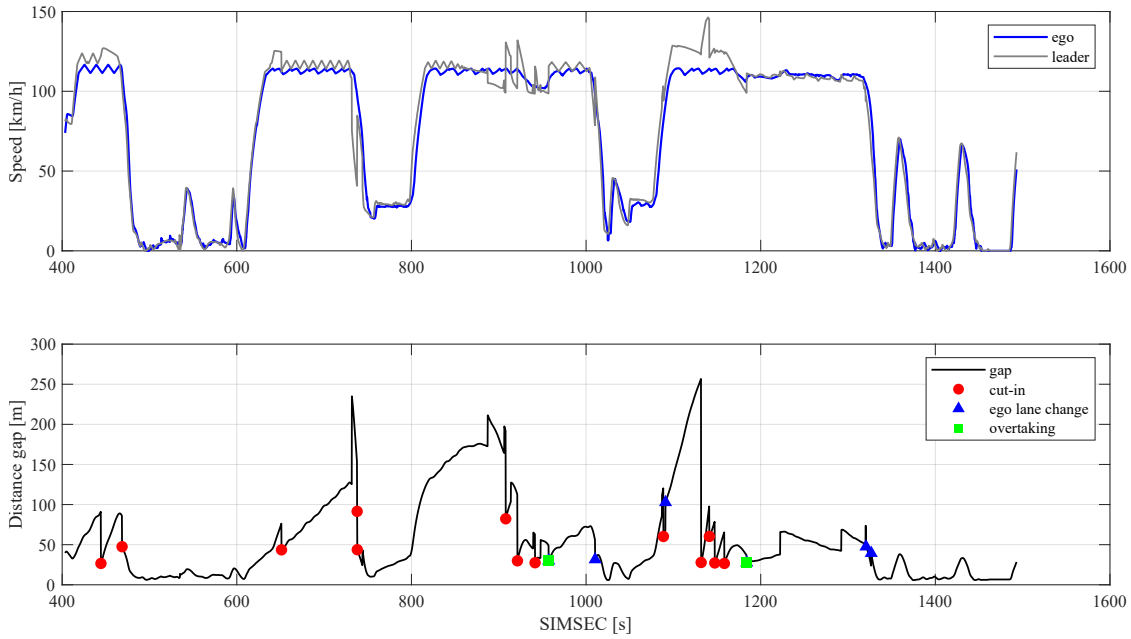


Figure 6: Overview of the optimized driving profile considering interactions between the ego and leading vehicles under a driving scenario

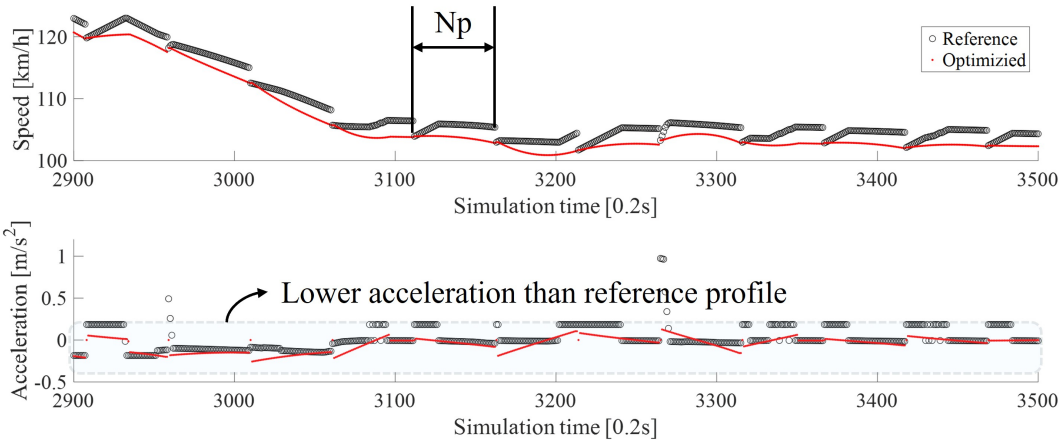


Figure 7: Speed and acceleration trajectory comparisons between reference and optimized profiles

of car-following characteristics, including desired time headway t_h, minimum spacing s_0, maximum acceleration a_{max}, and comfortable deceleration b [75, 76]. Here, s is the actual gap between the subject vehicle and the leading vehicle, s^* is the desired minimum gap defined as

$$
\begin{aligned}
a &= a_{max}\left[1 - \left(\frac{v}{v_0}\right)^{\delta} - \left(\frac{s^*}{s}\right)^2\right], \\
s^* &= s_0 + vt_h + \frac{v\Delta v}{2\sqrt{ab}},
\end{aligned}
\tag{31}
$$

where v_0 is the desired (free-flow) speed, δ is the acceleration exponent governing the sensitivity of acceleration to the speed ratio v/v_0, and $\Delta v = v - v_{lead}$ is the relative velocity (closing rate) between the subject and leading vehicle, with $\Delta v > 0$ indicating an approaching (gap-closing) condition.

For each car-following episode, the follower's speed and spacing trajectories were reconstructed by forward-simulating the IDM acceleration using the leading vehicle's observed speed profile as an exogenous input. The IDM parameters were calibrated by minimizing the residual between the simulated and observed speed and spacing trajectories across all episodes (described in the following section), using bounded nonlinear least-squares optimization in MATLAB R2024b. This trajectory-based calibration was adopted because per-timestep acceleration noise in the simulated traffic data precluded a stable parameter fit. The calibration results demonstrate that the IDM successfully reproduced the trajectory-level driving dynamics in terms of both vehicle speed and inter-vehicle spacing as shown in Figure 8a. Across all car-following episodes, the calibrated model achieved a speed RMSE of 1.6–1.8 m/s with an R^2 exceeding 0.95, while the gap RMSE ranged from 20 to 30 m with an R^2 of approximately 0.60. The larger spacing error mainly results from cumulative integration of small velocity discrepancies over long car-following episodes.

Among the calibrated parameters, the calibrated desired time headway t_h is the key parameter that determines driving behavioral characteristics. As illustrated in Figure 8b, most of the ego vehicle's calibrated t_h (ranging from 0.71 s to 2.51 s) was at or above the median t_h of the individually calibrated surrounding-vehicle population. This ego-versus-population gap remained stable once the full available vehicle population (628-1524 vehicles per case, rather than a small subsample) was used for calibration—the population median t_h converged to a narrow range of 0.73-0.84 s across all 15 cases—indicating that the observed difference reflects a systematic behavioral shift rather than sampling variability. Furthermore, Figure 8c independently supports this finding using directly measured headways. The median measured headway of the ego vehicle following the optimized profile is consistently higher than that of the surrounding traffic vehicles, confirming the tendency identified from the calibrated IDM parameters. These finding suggest that the optimized control strategy adopts a systematically larger following gap than typical car-following behavior, consistent with an eco-driving strategy that trades following distance for smoother, less reactive speed trajectories.

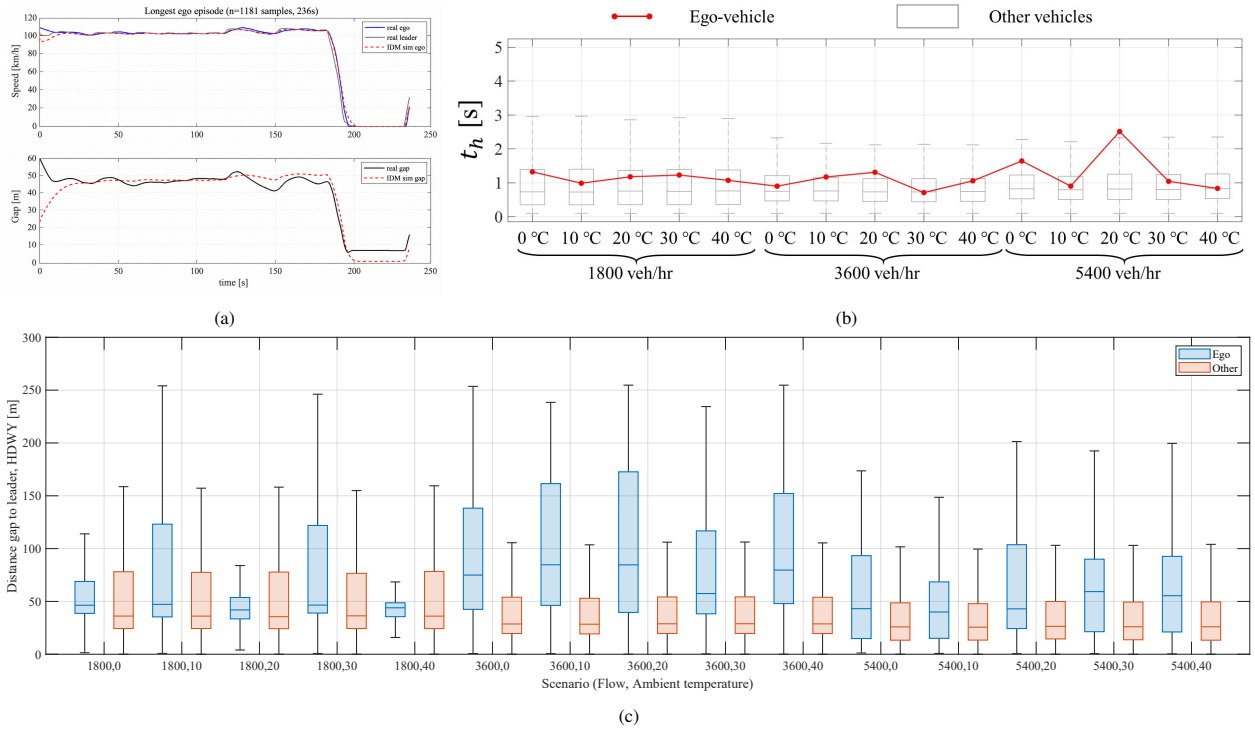


Figure 8: IDM-based driving behavioral analysis: (a) model fitting, (b) desired time headway analysis, (c) measured time headway distribution

Figure 9 further compares the battery current and aging cost between the reference and optimized driving profiles. The results show that the optimized profile consistently reduces the magnitude and variability of the battery current compared to the reference case. In particular, high current spikes observed in the reference profile are effectively mitigated under the optimized trajectory, resulting in smoother current demand over time. This reduction in current

fluctuation directly translates into a lower aging cost, as illustrated in the lower panel of Figure 9. The optimized profile exhibits significantly attenuated peaks in aging cost, especially during high-load operating conditions, indicating that aggressive electrochemical stress is alleviated. Overall, the results demonstrate that the proposed optimization framework not only improves energy efficiency through smoother driving behavior but also effectively suppresses battery degradation by limiting high-current events.

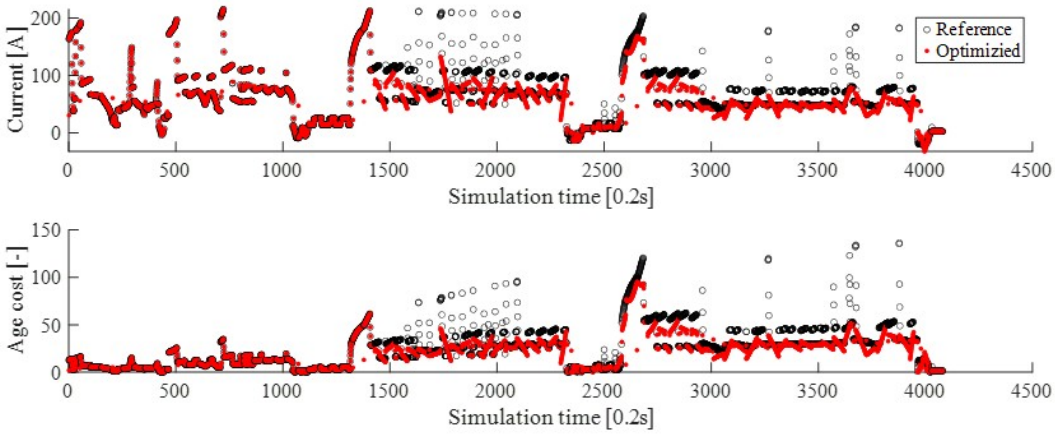


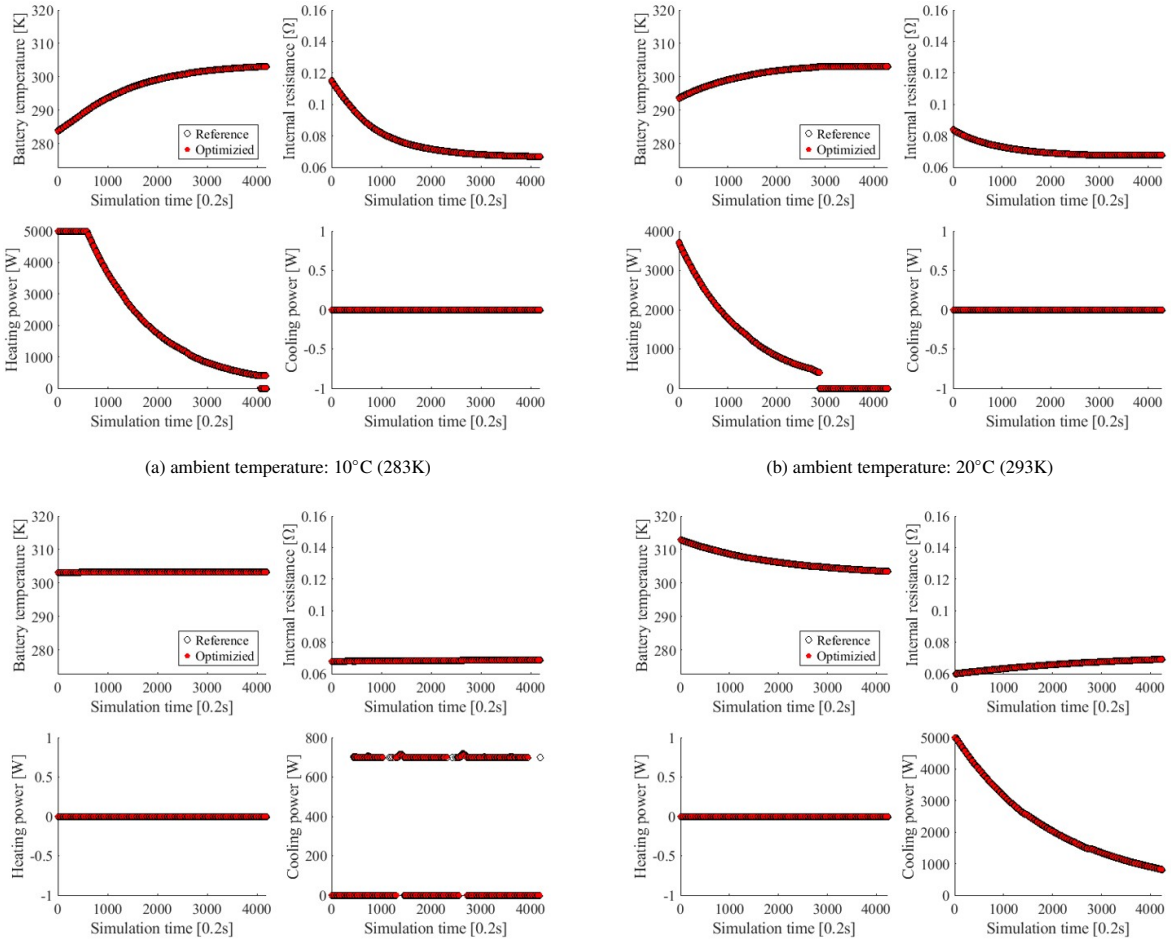
Figure 9: The comparisons of battery current and age cost in the objective function between reference and optimized profiles

On the other hand, Figure 10 shows that the BTMS operation remains identical for a given ambient temperature in the range of 10-40 °C. Specifically, the cooling and heating powers regulate the battery temperature toward a target value of 303 K. As the thermal control strategy is not included in the optimization solver, the BTMS operations (cooling and heating) are determined solely by the ambient temperature and remain unchanged regardless of the optimized driving profile. In this framework, heating power is activated when the battery temperature falls below the target value, whereas cooling power is activated when the temperature exceeds it. These control actions follow the rule-based strategy defined in Section 2.2, resulting in physically feasible BTMS operation for BEV systems. Therefore, the proposed optimization framework effectively captures BTMS operation as a function of the ambient temperature. Moreover, because internal resistance is temperature-dependent, its variation is implicitly incorporated into the optimization process, thereby enhancing the physical realism of the resulting solutions.

4.2. Comparison of optimization results under different conditions

This section compares the optimization margins across different ambient temperatures, traffic flows, and weighting factors. The optimization margins are defined as the relative reductions from the reference to the optimized profiles in terms of energy depletion and battery capacity loss, respectively. Table 3 summarizes the total optimization margins in each scenario between the reference and optimized driving profiles under different ambient temperatures and traffic flow conditions. Overall, the proposed optimization framework consistently improves both energy efficiency and battery degradation metrics across all scenarios with different weight factors (0, 0.999, 0.9999, and 1.0). In terms of driving behavior, the optimized profiles generally result in slightly lower average speeds compared to the reference. Despite this reduction, the total travel distance remains nearly unchanged, with differences within 400 m, indicating that the optimization primarily adjusts speed trajectories rather than altering trip completion. Detailed results for each operating condition and weighting factor are provided in Tables A1–A4 in the Appendix.

Across all ambient temperatures and traffic conditions, the optimized profiles exhibit reduced energy depletion, demonstrating enhanced energy efficiency. The optimization leads to a consistent reduction in battery capacity loss, as quantified by (19). The optimized cases show lower degradation values compared to the reference across all conditions, with particularly significant improvements observed under moderate-to-high temperature conditions (20–30°C). This behavior is consistent with findings reported in previous studies [13, 20, 77], indicating that the proposed speed control strategy can mitigate electrochemical stress by suppressing aggressive driving patterns.



(a) ambient temperature: 10°C (283K)

(b) ambient temperature: 20°C (293K)

(c) ambient temperature: 30°C (303K)

(d) ambient temperature: 40°C (313K)

Figure 10: Comparison of battery temperature, internal resistance, and BTMS operations between reference and optimized profile

The influence of ambient temperature can be attributed to its effects on battery operation. At lower temperatures, increased internal resistance leads to greater energy losses and higher heating demand to reach the target battery temperature, thereby limiting the achievable energy-saving benefit. As temperature increases, reduced internal resistance and improved battery efficiency provide more favorable conditions for energy reduction. At higher temperatures, however, increased cooling demand and temperature-dependent aging effects can limit further improvements in the optimization margin [6, 7].

On the other hand, no consistent trend in the optimization margins is observed with respect to traffic flow. Increasing traffic flow does not monotonically increase or decrease either the energy depletion or capacity loss margin, suggesting that traffic volume alone does not directly determine the effectiveness of predictive speed optimization. As discussed in the previous section, the optimized profiles are primarily achieved through smoother and less reactive driving behavior characterized by larger headways from preceding vehicles. Therefore, the achievable optimization margin is more closely associated with the resulting vehicle-level driving dynamics than with traffic volume itself. This observation motivates a more detailed analysis of speed-related driving characteristics and their interactions with environmental conditions in the following section.

These trends are also examined across different weighting factors. Figure 11 illustrates the relationship between the energy depletion and battery degradation margins as the weighting factor varies. It should be noted that the weighting factors are numerically concentrated near unity because the energy depletion and battery degradation terms have

Table 3: The comparisons of trip energy depletion margin (ζ_{Ah}^{trip}) and trip capacity loss margin (ζ_{loss}^{trip}) across the weight factors

Ambient temperature	Traffic flow	Weight λ							
		0		0.999		0.9999		1	
		ζ_{Ah}^{trip}	ζ_{loss}^{trip}	ζ_{Ah}^{trip}	ζ_{loss}^{trip}	ζ_{Ah}^{trip}	ζ_{loss}^{trip}	ζ_{Ah}^{trip}	ζ_{loss}^{trip}
0 °C	1800	8.0%	5.4%	6.6%	4.4%	4.7%	3.1%	2.1%	1.5%
	3600	8.1%	5.5%	6.2%	8.2%	5.3%	8.8%	1.7%	1.2%
	5400	8.6%	5.4%	7.5%	4.7%	4.6%	2.9%	2.9%	1.9%
10 °C	1800	10.6%	6.8%	9.3%	6.0%	6.0%	4.0%	3.8%	2.5%
	3600	10.4%	6.8%	9.2%	5.1%	6.8%	4.5%	2.8%	1.9%
	5400	11.1%	7.3%	9.2%	6.0%	6.6%	4.6%	1.7%	1.1%
20 °C	1800	13.0%	8.8%	11.3%	7.5%	8.8%	5.9%	4.8%	3.8%
	3600	14.0%	9.3%	11.9%	12.8%	8.3%	5.6%	3.6%	2.4%
	5400	12.7%	8.1%	10.6%	6.8%	7.8%	5.1%	3.0%	2.0%
30 °C	1800	14.2%	9.0%	12.5%	8.1%	7.6%	5.0%	5.1%	3.4%
	3600	13.7%	10.9%	12.5%	8.1%	8.0%	5.3%	3.8%	2.5%
	5400	13.3%	8.4%	12.6%	13.0%	7.6%	4.9%	2.0%	1.3%
40 °C	1800	12.6%	8.4%	10.0%	6.7%	8.5%	5.7%	4.4%	2.9%
	3600	12.3%	8.4%	11.6%	7.8%	8.1%	5.3%	2.6%	1.7%
	5400	12.7%	8.0%	10.8%	6.8%	7.9%	5.0%	3.0%	1.9%

substantially different numerical scales. Therefore, the absolute magnitude of λ should not be directly interpreted as the physical importance assigned to each objective. Instead, the weighting factors were selected to compensate for the scale difference between the two objective terms and to systematically examine the transition from degradation-oriented to energy-oriented optimization.

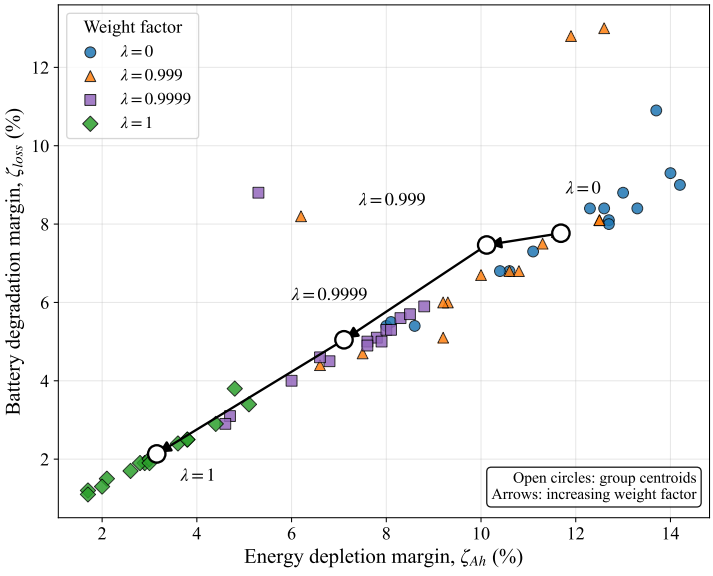


Figure 11: The shift of optimization margin regions with different weighting factors of 0, 0.999, 0.9999, and 1.0

Overall, the energy depletion and battery degradation margins exhibit a strong positive correlation, with a Pearson correlation coefficient (PCC) of 0.897, indicating that the two objectives are closely coupled through the underlying driving and battery operating conditions. This coupling can be attributed to their shared dependence on battery current demand governed by the resulting speed trajectory. Smoother acceleration and braking reduce high-current operation, thereby simultaneously affecting energy depletion and current-induced degradation stress. This relationship is con-

sistent with previous studies demonstrating that battery-degradation-aware speed optimization can also affect energy consumption [41, 42].

More importantly, the present results reveal that this coupling is asymmetric from an optimization perspective. As the weight assigned to battery degradation, $(1 - \lambda)$, decreases, both the energy depletion and degradation margins generally decrease. Notably, the energy depletion margin does not reach its maximum at $\lambda = 1$, where only the energy-related objective is considered. This result indicates that explicitly considering battery degradation can provide an additional energy-saving benefit beyond that achieved by energy-oriented optimization alone. Because energy depletion and battery degradation are coupled through the resulting battery operating conditions, degradation-aware optimization can favor operating conditions that are simultaneously beneficial to both objectives.

To further investigate the origin of this asymmetric cross-objective effect, Figure 12 compares the distributions of vehicle speed, acceleration, and battery C-rate under different weighting factors. While the median speed remains nearly unchanged across the weighting factors, the distributions of speed and acceleration become progressively wider as λ approaches unity. A similar tendency is observed for the battery C-rate, with a broader distribution and higher upper range under energy-oriented optimization. These results indicate that explicitly considering battery degradation tends to suppress dynamic driving fluctuations and high-load battery operation, whereas increasing the emphasis on energy depletion allows greater variations in vehicle dynamics and battery current while maintaining a comparable overall speed level.

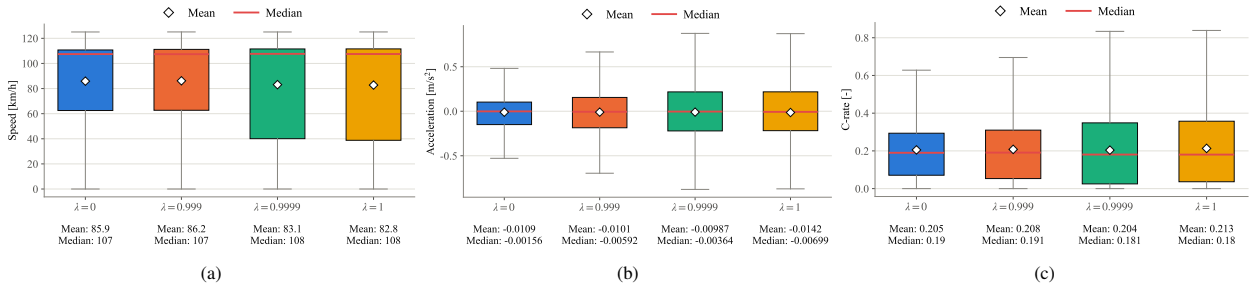


Figure 12: Effect of the weighting factor λ on the distributions of (a) vehicle speed, (b) acceleration, and (c) battery C-rate

The observed C-rate behavior provides a physical explanation for the asymmetric cross-objective effect. In the battery aging model employed in this study, C-rate is explicitly incorporated into the capacity-loss calculation in Eq. (19). Therefore, suppressing high-C-rate operation provides a direct mechanism for reducing battery degradation. At the same time, the associated smoother acceleration and braking behavior reduces high-current operation and can also contribute to lower energy depletion. Consequently, degradation-aware optimization promotes operating conditions that are favorable not only for battery health but also for energy efficiency.

In contrast, energy-oriented optimization primarily minimizes the energy-related objective and does not explicitly account for C-rate-dependent degradation stress. It may therefore permit larger variations in acceleration and battery current when such operation remains favorable from the energy perspective. This explains why increasing the emphasis on energy depletion does not provide a comparable degradation-reduction benefit and, importantly, why the largest energy depletion margin is not necessarily obtained at $\lambda = 1$. These findings demonstrate that the cross-objective benefits are asymmetric rather than reciprocal and highlight the importance of explicitly incorporating battery degradation into predictive speed optimization when simultaneous improvements in energy efficiency and battery health are desired.

4.3. Conditional effect analysis in optimization margins

In this section, we investigate the driving environmental conditions that lead to high optimization margins. Battery capacity loss is known to be strongly correlated with energy consumption [16, 19, 20, 78]; therefore, improvements in energy consumption can serve as a proxy for reductions in capacity loss. This relationship is further confirmed by Figure 11, where energy depletion and capacity loss margins exhibit a strong positive correlation around 0.9 of PCC. Accordingly, the subsequent analyses employ the energy depletion margin as the primary optimization metric for investigating the driving conditions associated with high optimization margins, while the corresponding capacity-loss

trends remain consistent. Hereafter, the optimization margin refers to the reduction in energy depletion achieved by the optimized driving profile relative to the reference profile.

The absolute optimization margin was used in the factor analysis because the objective was to identify the operating conditions under which a greater amount of energy can be saved over the same travel distance. Since all simulation cases were evaluated over an identical travel distance, differences in the absolute margin are not attributable to differences in trip length but reflect the combined effects of driving and environmental conditions on the achievable optimization benefit. Several vehicle-level parameters that are closely related to energy consumption have been widely reported in the literature. First, vehicle speed is recognized as one of the most influential factors affecting energy consumption among vehicle-level parameters [1, 2, 3, 4, 66, 51]. Second, road gradient has also been identified as a key factor influencing vehicle energy demand [79, 80, 81]. Third, battery temperature plays an important role due to its impact on internal resistance and thermal management requirements [5, 6, 7, 66, 82].

Building upon these findings, Figure 13 presents the relative influence of key variables on the optimization margin, quantified using the absolute PCC value. Among all variables, average speed exhibits the strongest correlation (approximately 0.68), indicating that it is the dominant factor governing the effectiveness of the optimization. This result suggests that high-speed operating conditions provide greater potential for optimization gains, particularly in terms of energy savings and degradation mitigation.

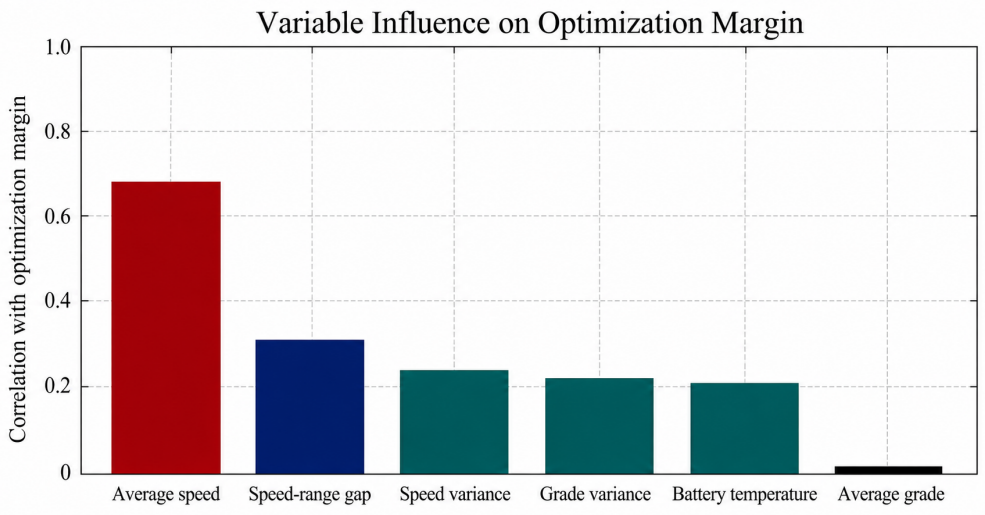


Figure 13: Correlation analysis between driving conditional parameters and optimization margin

The speed-range gap also shows a moderate correlation (around 0.32), implying that variations in allowable speed or speed constraints contribute to the optimization outcome, albeit to a lesser extent than the average speed. Similarly, speed variance and grade variance demonstrate comparable levels of influence (approximately 0.24 and 0.22, respectively), indicating that dynamic driving conditions—such as fluctuations in speed and road slope—play a secondary but non-negligible role in shaping optimization performance.

In contrast, battery temperature shows a relatively weak correlation (around 0.20), suggesting that its effect on optimization margin is indirect and less significant compared to driving-related factors. Notably, the average road grade exhibits a negligible correlation, indicating that the magnitude of slope itself is less critical than its variability. Overall, these results highlight that the optimization margin is primarily driven by speed-related variables, while environmental factors such as road gradient and temperature play a supporting role, mainly through their influence on dynamic driving conditions rather than static characteristics.

Based on the correlation analysis, a hierarchical investigation is conducted to identify the driving environmental conditions that lead to high optimization margins. Figure 14 illustrates the relationship between speed-related variables and the optimization margin. As shown in Figure 14a, the optimization margin increases with higher average speeds and larger speed-range gaps (i.e., speed min-max gaps), indicating that broader allowable speed control ranges under

high-speed conditions enhance the effectiveness of optimization.

Furthermore, Figure 14b shows that higher optimization margins are primarily observed in regions with high average speeds and moderate levels of speed variability. This suggests that dynamic yet controlled driving conditions—characterized by sufficient speed flexibility without excessive fluctuations—provide favorable conditions for achieving greater optimization benefits. Overall, these results confirm that speed-related factors play a dominant role in determining the magnitude of optimization margins. However, as shown in Figure 14b, there still exist data points exhibiting high optimization margins that are not located within regions of high average speed and low speed variability (see the gray dashed circle). To further explain this ambiguity, road gradient–related parameters are introduced as additional explanatory factors.

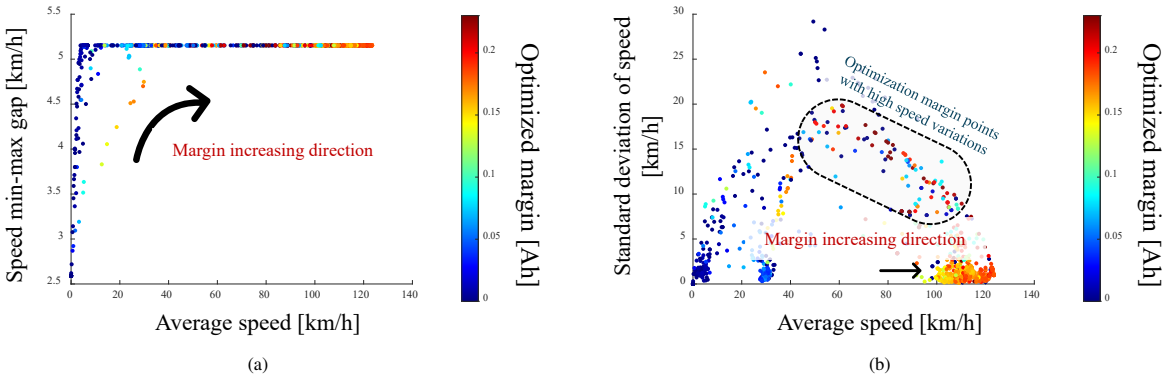


Figure 14: Trend of increasing energy consumption margin with respect to speed-related parameters in the scatter plot

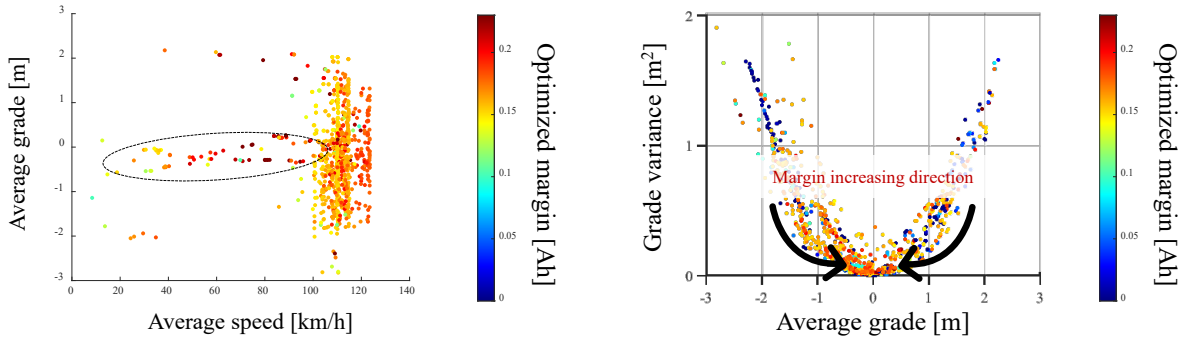
To further explain the remaining high-margin cases, road gradient–related parameters are investigated. Figure 15 illustrates the relationship between the average relative road grade (defined with respect to an initial reference grade), grade variability, and the optimization margin. Here, a positive average grade corresponds to an uphill segment, whereas a negative value represents a downhill segment. In addition, higher grade variance reflects more dynamic road gradient patterns over the optimization horizon.

Figure 15a illustrates the data points with optimization margins exceeding 0.1 Ah. The results indicate that high optimization margins are primarily observed in regions characterized by relatively low grade variance and average grades close to flat or mildly sloped conditions. In addition, high optimization margins are observed in both uphill and downhill segments, suggesting that optimization benefits are not limited to a specific slope direction. However, it highlights that, once favorable speed-related conditions are satisfied, stable road-gradient conditions provide greater opportunities for speed optimization by allowing smoother trajectory adjustment without excessive disturbance from rapidly varying slopes.

In particular, Figure 15b reveals a distinct nonlinear relationship between average grade and grade variance. The optimization margin tends to decrease as the magnitude of the average grade deviates from near-flat conditions, accompanied by higher grade variability. This indicates that road segments with frequent transitions between uphill and downhill—rather than uniformly flat or constant slopes—offer more favorable conditions for improving both energy efficiency and battery degradation through speed optimization. These results suggest that, beyond speed-related factors, the variability of road gradient plays a critical role in determining high optimization margins, especially in cases where speed conditions alone cannot fully explain the observed improvements.

Lastly, Figure 16 presents the influence of ambient temperature on the optimization margin across different speed ranges. As the ambient temperature increases from 0 to 40°C, the overall distribution of optimization margins shifts upward, indicating that higher temperatures generally provide more favorable conditions for optimization. At low temperatures (e.g., 0°C and 10°C), the optimization margin remains relatively limited across all speed ranges, and the variability is also constrained. This behavior can be attributed to increased internal resistance and higher heating demand, which reduce the flexibility of optimization and limit achievable performance improvements.

In contrast, at moderate temperatures (20–30°C), both the median and upper bound of the optimization margin significantly increase, particularly in higher speed regions. This suggests that thermally favorable conditions enhance



(a) (b)

Figure 15: Trend of increasing energy consumption margin with respect to additional grade-related parameters in the scatter plot

the effectiveness of speed optimization by reducing internal resistance and enabling more efficient energy utilization. Notably, the variability of the optimization margin also increases in this range, indicating that the potential for optimization becomes more sensitive to driving conditions. At higher temperatures (40°C), although the optimization margin remains relatively high, the rate of improvement tends to saturate compared to the 30°C case. This implies that excessive thermal conditions may introduce competing effects, such as increased cooling demand, which partially offsets the benefits gained from reduced internal resistance. Overall, these results indicate that the effectiveness of the proposed optimization framework is strongly dependent on the ambient temperature, with moderate temperature conditions providing the most favorable environment for achieving high optimization margins.

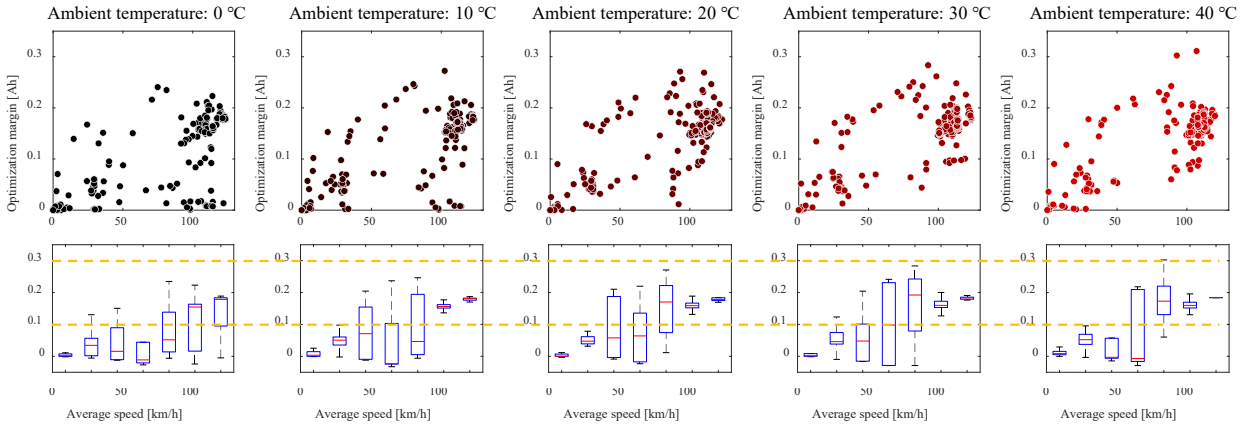


Figure 16: Trend of increasing energy consumption margin in scatter and box plots across different battery temperatures

The above observations suggest possible conditional effects among the dominant variables. Accordingly, interaction effects among the dominant variables were further examined using a multiple linear regression model as in (32). To facilitate direct comparison of the estimated coefficients and ensure numerical stability during parameter estimation, all explanatory variables (x) were standardized using z-score normalization (z_x) prior to regression analysis.

$$\zeta = \varepsilon_0 + \varepsilon_1 z_{T_{env}} + \varepsilon_2 z_{\theta_{var}} + \varepsilon_3 z_{\bar{v}} + \varepsilon_4 z_{v_{var}} + \varepsilon_5 z_{T_{env}} z_{\bar{v}} + \varepsilon_6 z_{\theta_{var}} z_{\bar{v}} + \varepsilon_7 z_{\bar{v}} z_{v_{var}} \tag{32}$$

where ζ denotes the optimization margin, T_{env} is the ambient temperature, θ_{var} represents the standard deviation of the relative road grade, $\bar{v}$ is the average vehicle speed, v_{var} denotes the standard deviation of vehicle speed, and ε_i are the regression coefficients estimated using robust linear regression. The interaction terms ($T_{env}\bar{v}$, $\theta_{var}\bar{v}$, and $\bar{v}v_{var}$) are introduced to quantify the conditional effects of the dominant driving and environmental variables identified from the

hierarchical analyses presented in Figure 14–16. The resulting regression model achieves an adjusted coefficient of determination (R^2_{adj}) of 0.944, indicating excellent explanatory capability for the optimization margin.

Table 4 summarizes the regression results for the interaction model. The results indicate that the interactions between average speed and road-grade variability, as well as between average speed and speed variability, are statistically significant ($p < 0.01$), whereas the interaction between ambient temperature and average speed is not significant. These findings suggest that the influence of average speed on the optimization margin is conditionally affected by road-gradient and speed variability, while the effect of ambient temperature is largely independent of the driving speed.

Table 4: Estimated coefficients of the interaction regression model

Variable	**Estimate**	**S.E.**	**t-statistic**	**p-value**
Intercept	1.106e-01	5.609e-04	197.170	< 0.001
Ambient temperature ($z_{T_{env}}$)	0.931e-03	4.341e-04	2.145	0.032
Grade variance ($z_{\theta_{var}}$)	-2.103e-03	6.717e-04	-3.130	0.002
Average speed ($z_{\bar{v}}$)	51.7e-03	7.083e-04	72.990	< 0.001
Speed variance ($z_{v_{var}}$)	-33.313e-03	6.449e-04	-51.653	< 0.001
Ambient temperature × Average speed	-0.0356e-03	4.399e-04	-0.081	0.936
Grade variance × Average speed	3.314e-03	1.022e-03	3.242	0.001
Average speed × Speed variance	-12.422e-03	6.614e-04	-18.781	< 0.001

The estimated coefficients demonstrate that average speed exerts the strongest positive influence on the optimization margin, whereas speed variability exhibits the largest negative effect among the examined variables. These findings indicate that predictive speed optimization is most effective under high-speed driving conditions with stable speed profiles, whereas frequent speed fluctuations reduce the optimization potential by limiting the ability to maintain smooth and energy-efficient vehicle trajectories.

The interaction terms further reveal that the influence of road-gradient variability and speed variability on the optimization margin depends on the average driving speed. Specifically, the positive interaction between average speed and road-gradient variability indicates that the adverse effect of varying road gradients becomes less pronounced as the average speed increases. In contrast, the negative interaction between average speed and speed variability suggests that maintaining a stable speed profile becomes increasingly important for maximizing the optimization margin under high-speed driving conditions.

Among the environmental factors, road-grade variability has the greatest influence on the optimization margin. Its negative coefficient indicates that lower road-grade variability is generally associated with larger optimization margins, suggesting that smoother terrain provides greater flexibility for predictive speed adjustment. Together with its significant interaction with average speed, this result demonstrates that the achievable optimization benefit is governed not only by individual driving or road conditions but also by their combined effects.

Ambient temperature, in comparison, has a relatively small contribution to the optimization margin. Although its main effect remains statistically significant ($p = 0.032$), it exhibits the largest p-value among the examined main effects. Moreover, the interaction between ambient temperature and average speed is not statistically significant ($p = 0.936$), suggesting that the temperature effect is primarily additive and does not vary appreciably with driving speed. It should also be noted that BTMS operation is determined by predefined control logic rather than directly optimized within the predictive speed control framework. Therefore, the relatively weak temperature dependence identified in this study should be interpreted within the thermal modeling scope of the present framework and does not necessarily imply that thermal conditions generally have a limited influence on predictive speed optimization.

In summary, the hierarchical and interaction analyses demonstrate that the effectiveness of predictive speed optimization is primarily governed by vehicle-level driving dynamics and their interaction with road conditions. High optimization margins are generally associated with high average speeds and stable speed profiles, while road-grade variability further conditions the achievable benefit depending on the driving-speed regime. In comparison, ambient temperature exhibits a smaller and largely additive contribution within the investigated thermal modeling conditions. These findings extend conventional optimization performance evaluation by identifying not only the individual factors associated with greater optimization benefits but also the combinations of operating conditions under which traffic-aware and degradation-aware predictive speed control can be most effective.

5. Conclusion

This study proposed a predictive speed control framework for battery electric vehicles (BEVs) that jointly considers energy consumption and battery degradation under realistic traffic conditions. To this end, an integrated MATLAB–VISSIM simulation framework was developed by coupling microscopic traffic simulation with vehicle energy conversion, battery thermal management system (BTMS) operation, and semi-empirical battery aging models. Based on this framework, optimal speed trajectories were determined under diverse traffic, road, and thermal conditions.

To quantitatively characterize optimization-induced changes in car-following behavior, the Intelligent Driver Model (IDM) was calibrated separately using the reference and optimized trajectories. The calibrated model successfully reproduced the overall trajectory-level dynamics, particularly the follower-speed profiles, while providing physically interpretable parameters associated with car-following behavior. The parameter comparison, together with the measured headway analysis, showed that the optimized trajectories generally maintain larger following headways and exhibit smoother and less reactive responses to surrounding traffic. These results indicate that the optimization benefits arise from systematic modifications of car-following behavior rather than solely from isolated reductions in instantaneous power demand.

The proposed framework consistently reduced both energy depletion and battery capacity loss across the examined operating conditions. The optimized trajectories suppressed aggressive acceleration and high-current events while maintaining comparable travel distances, thereby providing simultaneous benefits in energy efficiency and battery degradation. Furthermore, the analysis across different weighting factors revealed a strong positive correlation between the energy depletion and degradation margins, indicating that the two objectives are closely coupled through their shared dependence on driving dynamics and battery operating conditions. More importantly, this coupling exhibited an asymmetric cross-objective effect: degradation-aware optimization retained substantial energy-saving benefits, whereas energy-oriented optimization provided comparatively limited degradation-reduction benefits. This finding highlights the importance of explicitly accounting for battery degradation when designing predictive speed control strategies that seek simultaneous improvements in energy efficiency and battery health.

The analysis of driving and environmental factors further demonstrated that the achievable optimization margins are strongly condition-dependent. Speed-related variables were the dominant determinants of optimization effectiveness, with larger benefits generally observed under high average-speed and low speed-variability conditions. Road-grade variability also exhibited a significant conditional effect with average speed, indicating that the influence of terrain depends on the corresponding driving conditions. In comparison, ambient temperature showed a relatively smaller contribution to the optimization margins, although its influence remained statistically significant. These findings demonstrate that the effectiveness of predictive speed optimization is governed more strongly by vehicle-level driving dynamics and their interaction with road conditions than by individual environmental factors alone.

Overall, this study advances predictive speed control beyond aggregate performance evaluation by providing an integrated understanding of its performance, behavioral, and cross-objective characteristics. The findings demonstrate how predictive speed optimization modifies car-following behavior, reveal the coupled yet asymmetric relationship between energy efficiency and battery degradation, and identify the operating conditions under which the greatest benefits can be achieved. These insights provide a basis for developing traffic-aware and degradation-aware control strategies for future intelligent and electrified mobility systems. Future work will focus on real-time implementation under uncertain and dynamically changing traffic conditions, more detailed characterization of interactions with surrounding vehicles, and adaptive control using connected-vehicle information and real-world driving data.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist with grammar checking and language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgement

This work was supported by the InnoCORE program of the Ministry of Science and ICT(N10260002).

6. Appendix

To further examine the sensitivity of the optimization results to the weighting factor, Tables A1–A4 present detailed comparisons between the reference (Ref.) and optimized (Opt.) driving profiles for different weighting factors (λ = 0, 0.999, 0.9999, and 1). The results include average speed, energy depletion, battery capacity loss, and travel distance under different ambient temperatures and traffic flow conditions.

Table A1: The comparisons between reference (Ref.) and optimization (Opt.) in BEVs with the weight factor of 0

Ambient temperature	Flow	Speed [km/h]		Energy depletion [Ah]		Capacity loss [%]		Distance [km]	
		Ref.	Opt.	Ref.	Opt.	Ref.	Opt.	Ref.	Opt.
0 °C	1800 vph	97.7	96.6	15.1	13.9 (8.0%↑)	0.0544	0.0515 (5.4%↑)	22.2	22.0
	3600 vph	92.4	91.4	14.1	12.9 (8.1%↑)	0.0528	0.0499 (5.5%↑)	22.1	21.8
	5400 vph	74.0	73.1	15.3	14.0 (8.6%↑)	0.0571	0.0540 (5.4%↑)	22.2	21.9
10 °C	1800 vph	97.8	96.5	14.6	13.0 (10.6%↑)	0.0558	0.0520 (6.8%↑)	22.3	22.0
	3600 vph	90.6	89.3	13.5	12.1 (10.4%↑)	0.0549	0.0512 (6.8%↑)	22.1	21.8
	5400 vph	74.1	73.1	14.7	13.1 (11.1%↑)	0.0565	0.0524 (7.3%↑)	22.2	21.9
20 °C	1800 vph	96.8	95.2	13.3	11.6 (13.0%↑)	0.0537	0.0490 (8.8%↑)	22.3	22.0
	3600 vph	83.5	82.1	12.5	10.8 (14.0%↑)	0.0518	0.0470 (9.3%↑)	22.3	21.9
	5400 vph	82.4	81.2	14.0	12.2 (12.7%↑)	0.0554	0.0509 (8.1%↑)	22.2	21.9
30 °C	1800 vph	97.9	96.5	13.4	11.5 (14.2%↑)	0.0542	0.0493 (9.0%↑)	22.3	22.0
	3600 vph	90.5	89.1	12.0	10.4 (13.7%↑)	0.0550	0.0490 (10.9%↑)	22.1	21.8
	5400 vph	82.2	81.1	13.2	11.4 (13.3%↑)	0.0540	0.0495 (8.4%↑)	22.2	21.9
40 °C	1800 vph	96.7	95.1	13.9	12.2 (12.6%↑)	0.0558	0.0511 (8.4%↑)	22.3	21.9
	3600 vph	90.9	89.4	13.6	12.0 (12.3%↑)	0.0561	0.0514 (8.4%↑)	22.2	21.8
	5400 vph	82.4	81.0	14.7	12.8 (12.7%↑)	0.0565	0.0520 (8.0%↑)	22.2	21.8

Table A2: The comparisons between reference and optimization in BEVs with the weight factor of 0.999

Ambient temperature	Flow	Speed [km/h]		Energy depletion [Ah]		Capacity loss [%]		Distance [km]	
		Ref.	Opt.	Ref.	Opt.	Ref.	Opt.	Ref.	Opt.
0 °C	1800 vph	97.3	96.7	14.8	13.9 (6.6%↑)	0.0537	0.0513 (4.4%↑)	22.2	22.0
	3600 vph	91.0	90.4	14.0	13.1 (6.2%↑)	0.0550	0.0505 (8.2%↑)	22.0	21.8
	5400 vph	73.7	73.1	15.1	14.0 (7.5%↑)	0.0569	0.0542 (4.7%↑)	22.1	21.9
10 °C	1800 vph	97.4	96.7	14.5	13.1 (9.3%↑)	0.0555	0.0522 (6.0%↑)	22.2	22.0
	3600 vph	90.1	89.3	13.2	12.0 (9.2%↑)	0.0535	0.0508 (5.1%↑)	22.0	21.8
	5400 vph	74.6	73.8	14.5	13.2 (9.2%↑)	0.0559	0.0526 (6.0%↑)	22.2	21.9
20 °C	1800 vph	97.6	96.6	13.9	12.3 (11.3%↑)	0.0548	0.0507 (7.5%↑)	22.2	22.0
	3600 vph	91.6	90.3	12.8	11.2 (11.9%↑)	0.0558	0.0487 (12.8%↑)	22.1	21.8
	5400 vph	81.9	81.1	13.6	12.1 (10.6%↑)	0.0507	0.0507 (6.8%↑)	22.1	21.9
30 °C	1800 vph	97.6	96.5	13.2	11.5 (12.5%↑)	0.0537	0.0494 (8.1%↑)	22.2	22.0
	3600 vph	85.4	84.3	12.0	10.5 (12.5%↑)	0.0512	0.0471 (8.1%↑)	22.3	22.0
	5400 vph	80.2	79.4	12.8	11.2 (12.6%↑)	0.0562	0.0489 (13.0%↑)	22.1	21.8
40 °C	1800 vph	96.2	95.2	13.8	12.4 (10.0%↑)	0.0556	0.0519 (6.7%↑)	22.2	22.0
	3600 vph	91.5	90.2	13.5	11.9 (11.6%↑)	0.0544	0.0502 (7.8%↑)	22.1	21.8
	5400 vph	82.2	81.1	14.6	13.0 (10.8%↑)	0.0561	0.0523 (6.8%↑)	22.2	21.9

Overall, the optimized trajectories reduce both energy depletion and battery capacity loss across the examined operating conditions while maintaining comparable average speeds and travel distances. However, the magnitude

Table A3: The comparisons between reference and optimization in BEVs with the weight factor of 0.9999

Ambient temperature	Flow	Speed [km/h]		Energy depletion [Ah]		Capacity loss [%]		Distance [km]	
		Ref.	Opt.	Ref.	Opt.	Ref.	Opt.	Ref.	Opt.
0 °C	1800 vph	96.8	96.6	14.4	13.7 (4.7%↑)	0.0527	0.0511 (3.1%↑)	22.0	22.0
	3600 vph	90.6	90.3	13.9	13.1 (5.3%↑)	0.0554	0.0505 (8.8%↑)	21.9	21.8
	5400 vph	74.2	74.0	15.2	14.5 (4.6%↑)	0.0568	0.0551 (2.9%↑)	22.1	22.0
10 °C	1800 vph	96.8	96.5	14.1	13.2 (6.0%↑)	0.0546	0.0524 (4.0%↑)	22.1	22.0
	3600 vph	84.6	84.2	13.6	12.6 (6.8%↑)	0.0536	0.0512 (4.5%↑)	22.1	22.0
	5400 vph	71.8	71.5	14.1	13.1 (6.6%↑)	0.0569	0.0542 (4.6%↑)	21.9	21.9
20 °C	1800 vph	95.6	95.3	12.7	11.6 (8.8%↑)	0.0521	0.0490 (5.9%↑)	22.0	22.0
	3600 vph	90.0	89.5	13.0	11.9 (8.3%↑)	0.0531	0.0501 (5.6%↑)	22.0	21.9
	5400 vph	73.3	73.1	13.6	12.5 (7.8%↑)	0.0546	0.0518 (5.1%↑)	22.0	21.9
30 °C	1800 vph	95.5	95.2	11.9	11.0 (7.6%↑)	0.0507	0.0481 (5.0%↑)	22.0	21.9
	3600 vph	83.6	83.3	11.7	10.8 (8.0%↑)	0.0505	0.0479 (5.3%↑)	22.1	22.0
	5400 vph	71.7	71.5	12.6	11.7 (7.6%↑)	0.0529	0.0503 (4.9%↑)	21.9	21.8
40 °C	1800 vph	95.7	95.3	13.9	12.7 (8.5%↑)	0.0559	0.0527 (5.7%↑)	22.1	22.0
	3600 vph	84.7	84.3	13.7	12.6 (8.1%↑)	0.0543	0.0515 (5.3%↑)	22.1	22.0
	5400 vph	73.4	73.1	14.3	13.2 (7.9%↑)	0.0557	0.0529 (5.0%↑)	22.0	21.9

Table A4: The comparisons between reference and optimization in BEVs with the weight factor of 1.0

Ambient temperature	Flow	Speed [km/h]		Energy depletion [Ah]		Capacity loss [%]		Distance [km]	
		Ref.	Opt.	Ref.	Opt.	Ref.	Opt.	Ref.	Opt.
0 °C	1800 vph	96.7	96.6	14.4	14.1 (2.1%↑)	0.0527	0.0519 (1.5%↑)	22.0	22.0
	3600 vph	84.5	84.3	14.3	14.0 (1.7%↑)	0.0542	0.0536 (1.2%↑)	22.1	22.0
	5400 vph	73.2	73.1	15.0	14.5 (2.9%↑)	0.0564	0.0554 (1.9%↑)	22.0	21.9
10 °C	1800 vph	96.7	96.5	14.0	13.5 (3.8%↑)	0.0545	0.0531 (2.5%↑)	22.0	22.0
	3600 vph	85.4	85.3	13.4	13.0 (2.8%↑)	0.0532	0.0522 (1.9%↑)	22.1	22.0
	5400 vph	73.1	73.0	14.1	13.8 (1.7%↑)	0.0550	0.0544 (1.1%↑)	21.9	21.9
20 °C	1800 vph	94.3	94.1	12.6	12.0 (4.8%↑)	0.0534	0.0514 (3.8%↑)	22.0	22.0
	3600 vph	84.5	84.3	12.5	12.1 (3.6%↑)	0.0519	0.0507 (2.4%↑)	22.1	22.0
	5400 vph	72.4	72.2	13.5	13.1 (3.0%↑)	0.0544	0.0533 (2.0%↑)	21.9	21.9
30 °C	1800 vph	96.9	96.7	12.9	12.3 (5.1%↑)	0.0531	0.0513 (3.4%↑)	22.1	22.0
	3600 vph	85.5	85.3	12.4	11.9 (3.8%↑)	0.0521	0.0508 (2.5%↑)	22.1	22.0
	5400 vph	73.1	73.0	12.9	12.6 (2.0%↑)	0.0534	0.0528 (1.3%↑)	21.9	21.9
40 °C	1800 vph	95.5	95.3	13.4	12.9 (4.4%↑)	0.0546	0.0530 (2.9%↑)	22.0	22.0
	3600 vph	84.5	84.3	13.8	13.4 (2.6%↑)	0.0545	0.0535 (1.7%↑)	22.1	22.0
	5400 vph	74.9	74.8	14.3	13.9 (3.0%↑)	0.0557	0.0546 (1.9%↑)	22.1	22.0

of these improvements varies with the weighting factor, reflecting the influence of the relative emphasis placed on energy consumption and battery degradation in the objective function. It should be noted that the weighting factors are concentrated near unity because of the substantial difference in numerical scale between the energy depletion and battery degradation terms; therefore, their absolute values should not be directly interpreted as the relative physical importance of the two objectives. These additional results support the weighting-factor-dependent trends discussed in the main text.

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